

Falconer's distance problem for the gauges $r e^{-(\log 1/r)^\theta}$,

$$\theta > 2/3$$

a complete proof with scale-dependent parameters

Notes for the *Falconer* project

October 8, 2026

Abstract

Let $\frac{2}{3} < \theta < 1$ and $h_\theta(r) = r \exp(-(\log 1/r)^\theta)$. We prove that every compact set $E \subset \mathbb{R}^2$ with $\mathcal{H}^{h_\theta}(E) > 0$ has a distance set of positive length. Such sets can have Hausdorff dimension exactly 1, so this goes below every power gauge $r^{1+\delta}$. The proof runs the planar argument of OpenAI's preprint [10], in the organization of the notes [7], with parameters that shrink with the scale $R = 2^N$: the gain is $\beta_N \approx N^{\theta-1}$, the block length is $\kappa_N \approx \beta_N$, and the tolerance is $\varepsilon_N \approx \beta_N^2$. Every estimate is proved with explicit dependence on these parameters, including the stationary-phase and integration-by-parts steps that [7] only sketched. Three ingredients replace fixed-parameter steps: an Orlicz estimate for radial projections and a summable reconstruction criterion, both following the research note [9], and a regular decomposition into $2^{O(1/\varepsilon)}$ pieces in which pieces lighter than $R^{-\kappa/2}$ are discarded. Three costs of size $2^{O(1/\varepsilon)}$ or $N^{O(1/\varepsilon)}$ (the number of pieces, the constants of the low-frequency steps, and the number of branches of the induction) must be paid by R^{κ_N} . This requires $\kappa_N^3 N \gg \log N$, and since $\kappa_N^3 N \approx N^{3\theta-2}$, this is exactly where $\theta > 2/3$ is used.

Contents

1	Statement and overview	1
1.1	How the proof goes	2
1.2	What is new compared with [7]	3
2	Conventions and parameters	3
2.1	Conventions	3
2.2	The parameters	4
3	Analytic tools with explicit constants	5
3.1	A bump function with explicit derivatives	5
3.2	Non-stationary phase	6
3.3	Stationary phase on the circle	7
3.4	Separation of variables	10
4	Distance measures and a summable criterion	10

5	Preparation	11
5.1	Frostman's lemma for the gauge	11
5.2	Orlicz estimates for radial projections	11
5.3	Two prepared pieces	13
5.4	Regular pieces	14
5.5	The excess function	15
6	The filter	16
6.1	Two tests	16
6.2	Masks and levels	17
6.3	Symbols	18
6.4	The filtered measure	19
7	Two energies and four estimates	19
7.1	The two energies	20
7.2	Estimate I: Mattila's identity with masks	21
7.3	Estimate II: orthogonality	23
7.4	Estimate III: linearization	25
7.5	Estimate IV: splitting in space	26
8	The induction for one piece	28
8.1	Budget and schedule	28
8.2	States and moves	29
8.3	The budget pays the costs	30
8.4	Chains are short	30
8.5	The energy decay theorem	31
9	Proof of Theorem 1.1	31
9.1	The entry depth and the lists	31
9.2	The shell estimate	32
9.3	Conclusion	33
10	Remarks	33
A	Ledger of the scale-dependent conditions	35

1 Statement and overview

Theorem 1.1. *Let $\frac{2}{3} < \theta < 1$, and let*

$$h_\theta(r) = r \exp\left(-(\log(1/r))^\theta\right) \quad (0 < r < 1), \quad h_\theta(0) = 0.$$

If $E \subset \mathbb{R}^2$ is compact and $\mathcal{H}^{h_\theta}(E) > 0$, then $\Delta(E) = \{|x - y| : x, y \in E\}$ has positive Lebesgue measure.

In particular the theorem covers every $\theta > 3/4$, the range first aimed at.

Remark 1.2 (about the gauge).

- (a) With $\ell = \log(1/r)$, $\frac{d}{dr} h_\theta(r) = e^{-\ell^\theta} (1 + \theta \ell^{\theta-1}) > 0$, so h_θ is increasing on $(0, 1)$. It is doubling near 0: by the mean value theorem, $\ell^\theta - (\ell - \log 2)^\theta \leq \theta \log 2$ when $\ell - \log 2 \geq 1$, so

$$h_\theta(2r) \leq 4 h_\theta(r) \quad \text{for } 0 < r \leq 1/(2e). \quad (1.1)$$

- (b) For every $\delta > 0$, $r^{1+\delta} \leq h_\theta(r) \leq r$ for small r , and $h_\theta(r)/r^{1+\delta} \rightarrow \infty$. Hence $\dim_H E > 1$ implies $\mathcal{H}^{h_\theta}(E) > 0$, and $\mathcal{H}^{h_\theta}(E) > 0$ implies $\mathcal{H}^1(E) = \infty$. A set with $0 < \mathcal{H}^{h_\theta}(E) < \infty$ (Cantor-type constructions give such compact sets) has $\mathcal{H}^{1+\delta}(E) = 0$ for all $\delta > 0$, so its Hausdorff dimension is exactly 1.
- (c) If h is any gauge with $h(r) \leq C h_\theta(r)$ for small r and some $\theta \in (2/3, 1)$, then $\mathcal{H}^h(E) \leq C \mathcal{H}^{h_\theta}(E)$, so the conclusion of Theorem 1.1 holds for h as well. In particular the theorem for one θ implies it for every larger θ .

1.1 How the proof goes

The proof is the planar argument of [10], as reorganized in [7], run at one scale $R = 2^N$ at a time with parameters that depend on N . We recall the shape of that argument and say what changes.

For two pieces μ_1, μ_2 of a Frostman measure on E , consider their distance measure α . At each scale R one removes from α the pairs (x, y) whose direction is bad, as seen from x or from y , and obtains a measure $\tau_N \leq \alpha$. One shows two things.

- *Little is removed.* The bad directions at each point form a set of small arc length, and the other piece cannot concentrate in them because its radial projections are spread out.
- *What remains has small energy at frequency R .* An induction over scales, driven by four estimates and paid for by a budget, bounds the Fourier energy of the remaining measure by $R^{-\beta}$, where β is the gain.

A criterion for absolute continuity then turns the two facts, for all N , into $|\Delta(E)| > 0$.

With $\dim_H E > 1 + \delta$ all parameters are fixed: $\beta = \delta/10$, and the block length κ and the tolerance ε are small constants. Every step of the induction loses a factor $R^{C\varepsilon}$, there are $O(1/\kappa)$ steps, and the losses $R^{C\kappa}$, $R^{C\varepsilon/\kappa}$ are beaten by the gain R^β .

For the gauge h_θ the linear density at scale 2^{-n} is $\exp(-(n \log 2)^\theta)$, so the gain available at scale $R = 2^N$ is only

$$R^{\beta_N} = 2^{cN^\theta}, \quad \beta_N \approx N^{\theta-1} \rightarrow 0.$$

So κ and ε must also tend to 0. We take $\kappa_N \approx \beta_N$ and $\varepsilon_N \approx \kappa_N \beta_N \approx \beta_N^2 \approx N^{2\theta-2}$. Every statement that [7] proves “for N large depending on κ, ε ” must then be checked again, with explicit constants. This is what these notes do. The constraints that come out are listed in Lemma 2.2 and Appendix A.

Where 2/3 comes from. Three costs of size $2^{O(1/\varepsilon)}$ or $N^{O(1/\varepsilon)}$ appear: the regular decomposition cuts each piece into $2^{O(1/\varepsilon)}$ parts (Lemma 5.8); a chain of the induction can contain $O(1/\varepsilon)$ low-frequency steps, each costing an absolute constant; and the induction tree has $N^{O(1/\kappa+1/\varepsilon)}$ branches. All three must be paid by $R^\kappa = 2^{\kappa N}$, a loss the budget can absorb. This needs $\varepsilon^{-1} \log N \lesssim \kappa N$. With $\varepsilon \approx \kappa^2$ and $\kappa \approx N^{\theta-1}$ this is $N^{3\theta-2} \gtrsim \log N$, so $\theta > 2/3$. Every other condition holds for $\theta > 9/16$ (see §10).

Remark 1.3 (on 3/4). A first version of this argument needed $\theta > 3/4$, because it discarded the regular parts of mass below $R^{-\varepsilon}$; the discarded mass $2^{O(1/\varepsilon)} R^{-\varepsilon}$ is then small only if $\varepsilon^2 N \gtrsim 1$, that is $N^{4\theta-3} \gtrsim 1$. The mass of a part enters the proof only through the lower bound for the excess function (Lemma 5.10(iii)), where a loss of order κ is affordable. So the light parts are discarded at mass $R^{-\kappa/2}$ instead, and 3/4 improves to 2/3.

1.2 What is new compared with [7]

1. **Radial projections.** Orponen’s L^s estimate [8] needs an energy $I_{1+\delta/2}$ that is infinite for sets of dimension 1. It is replaced by an Orlicz estimate (Theorem 5.4): the radial projections have finite $\int g \log^8(e + g)$. We follow the argument of [9] and give it in full. With it, the filter removes mass at most $C(\varepsilon N)^{-8} + CN R^{-\varepsilon} + CR^{-\kappa/4}$ at scale N , which is summable in N .
2. **Criterion.** Absolute continuity follows from summability of the removed masses and of the shell energies (Lemma 4.2, as in [9, Lemma 7]). No power decay is needed for the removed mass.
3. **Regular decomposition.** Proved in full, with explicit bounds, and with light parts discarded at mass $R^{-\kappa/2}$ (Lemma 5.8).
4. **Oscillatory estimates.** The four estimates are proved in full, with explicit constants. “Negligible” now means at most R^{-20} ; this needs about $1/\varepsilon$ integrations by parts, so all derivative bounds are tracked up to order $O(T)$, $T \approx 240/\varepsilon$. The masks are built from an explicit bump with $\|\omega^{(k)}\|_{L^1} \leq (\pi^2/6)^k (k!)^2$, and stationary phase on the circle is done with an explicit remainder (§3).
5. **Entry depth.** The excess function now satisfies $A(n) \geq g(n) - \kappa$ with $g(n) \approx n^\theta/N$ (Lemma 5.10), and the entry depth is chosen with respect to β_N (Lemma 9.2).

The tests, masks, budget, schedule, moves and the bound on chain length are as in [7], with constants made explicit. Their proofs are repeated here so that the document is self-contained.

Remark 1.4 (why the Lean formalization cannot be cited). The Lean statements in [5] have the form “for fixed parameters there is N_0 such that for all $N \geq N_0 \dots$ ” (for example `source_profile_equation25` and `exists.bounded.regularization.control`). With parameters depending on N , the threshold N_0 could exceed N , so these statements say nothing at scale N . This is why every estimate is reproved here with explicit dependence on the parameters. The two constraints that [9, §7.2] extracts from the Lean source are, first, $\lambda N > \sqrt{N}$, where one parameter λ serves both as the regularization tolerance and as the threshold for discarding parts, and, second, a recursive loss $(500\kappa + 10^6(\lambda + \sigma)/\kappa + \eta)N$ in the exponent. Here the two roles of λ are separated: the tolerance is ε and the discarding threshold is $R^{-\kappa/2}$. The second constraint corresponds to our choices $\kappa_N \lesssim \beta_N$ and $\varepsilon_N \lesssim \kappa_N \beta_N$.

2 Conventions and parameters

2.1 Conventions

- $\log$ is the natural logarithm and $\log_2$ the logarithm to base 2. The letter e denotes Euler's number.
- $B(x, r)$ is the closed disc of radius r about x . S^1 is the unit circle with arc length dw . We write $w = (\cos \phi, \sin \phi)$, and for a function f on S^1 we write $\partial_w^k f$ for the k -th derivative in ϕ . For $x \neq y$, $e(x, y) = (x - y)/|x - y|$ is the direction of the segment from y to x .
- The Fourier transform of a finite measure μ on $\mathbb{R}^d$ ($d = 1, 2$) is $\hat{\mu}(\xi) = \int e^{-i\xi \cdot x} d\mu(x)$, and similarly for functions.
- **Scales and cells.** N is a large integer and $R = 2^N$. A *depth* is an integer $n \in \{0, \dots, N\}$; it stands for the length 2^{-n} . An n -*cell* is a half-open dyadic square of side 2^{-n} ; c_X denotes the center of the cell X . Given a measure, a cell is *occupied* if it has positive mass. Every n -cell is the union of its four *children*, which are $(n + 1)$ -cells. An n -cell lies in the disc of radius $2^{-n}/\sqrt{2}$ about its center.
- **Constants.** An *absolute constant* is a number that depends on nothing (not on N , θ , E , nor on any cell, direction or frequency). Absolute constants are written explicitly, or, in a few places, as C with a bound stated nearby; all of them are below 2^{90} . The symbol N_0 denotes a threshold that depends only on θ and on the set E (through the Frostman constant C_0 of Proposition 5.6). All statements about scale N are claimed for $N \geq N_0$.
- A quantity is *negligible* if it is at most R^{-20} .

2.2 The parameters

Fix $\theta \in (2/3, 1)$. For an integer $N \geq 4$ put

$$g(n) = \frac{(n \log 2)^\theta}{N \log 2} \quad (0 \leq n \leq N), \quad \text{so that} \quad R^{-g(n)} = e^{-(n \log 2)^\theta} = 2^n h_\theta(2^{-n}) \quad (n \geq 1). \quad (2.1)$$

The function g is increasing and concave, $g(0) = 0$, and $g(N) = (N \log 2)^{\theta-1}$.

Definition 2.1 (parameters).

$$\begin{aligned} \beta &= \tfrac{1}{2} g(N/4) && \text{the gain;} \\ \kappa &= \lceil \beta N / 1000 \rceil / N, \quad q = \kappa N && \text{the block length;} \\ \varepsilon &= \lceil \kappa^2 N / 10^6 \rceil / N && \text{the tolerance;} \\ T &= \lceil 240 / \varepsilon \rceil && \text{the number of terms in expansions and of integrations by parts;} \\ I_* &= \lceil 7 / \kappa + 1 / \varepsilon \rceil + 3 && \text{the number of levels of masks.} \end{aligned}$$

So κN , q and εN are integers. Explicitly $\beta = c_\theta N^{\theta-1}$ with $c_\theta = \frac{1}{2}(\log 2)^{\theta-1} 4^{-\theta} \in [\frac{1}{8}, \frac{1}{4}]$. Hence, as $N \rightarrow \infty$,

$$\beta \asymp \kappa \asymp N^{\theta-1}, \quad \varepsilon \asymp N^{2\theta-2}, \quad \varepsilon N \asymp N^{2\theta-1}, \quad \kappa^3 N \asymp N^{3\theta-2}, \quad T \asymp I_* \asymp N^{2-2\theta},$$

with implied constants depending only on θ .

Lemma 2.2 (parameter facts). *There is $N_{\text{par}}(\theta)$ such that for $N \geq N_{\text{par}}(\theta)$:*

$$(P1) \quad 2^{80}(T+1)^{12}(N+16)^{12} \leq R^{\varepsilon/8};$$

$$(P2) \quad \varepsilon N \geq 16, \text{ and } \log_2(49/\varepsilon) + 3.33(8/\varepsilon + 1) \leq \kappa N/4;$$

$$(P3) \quad (7/\kappa + 1/\varepsilon + 2) \log_2(N+16) \leq \kappa N/4, \text{ and } \log_2(N+16) \geq 10;$$

$$(P4) \quad \beta/1000 \leq \kappa \leq \beta/500 \leq 10^{-4}, \quad \kappa^2/10^6 \leq \varepsilon \leq \kappa^2/(5 \cdot 10^5), \quad 2/N \leq \beta/100, \quad I_* \leq T, \quad \text{and} \\ 2(1/\kappa + 2)(N+1) \leq N^2.$$

Proof. All quantities are explicit powers of N up to constants depending on θ . (P1): the logarithm of the left side is $O(\log N)$, while $\varepsilon N/8 \asymp N^{2\theta-1}$ and $2\theta - 1 > 0$. (P2): $\varepsilon N \rightarrow \infty$; and the left side of the second inequality is $\asymp 1/\varepsilon \asymp \kappa^{-2}$, while the right side is $\asymp \kappa N$, so it holds once $\kappa^3 N \asymp N^{3\theta-2}$ is large. (P3): the left side is $\asymp \kappa^{-2} \log N$ and the right side is $\asymp \kappa N$; it holds once $\kappa^3 N \gg \log N$, that is, for large N since $3\theta - 2 > 0$. (P4): $\kappa \leq \beta/1000 + 1/N$ and $1/N = o(\beta)$; $\beta/500 \leq 10^{-4}$ because $\beta \rightarrow 0$; $\varepsilon \leq \kappa^2/10^6 + 1/N$ and $1/N = o(\kappa^2)$ because $\kappa^2 \asymp N^{2\theta-2}$ and $2\theta - 2 > -1$; $2/N = o(\beta)$; $I_* \leq 7/\kappa + 1/\varepsilon + 4 \leq 2/\varepsilon \leq T$ because $1/\kappa = o(1/\varepsilon)$; and $1/\kappa = o(N)$. The condition $\theta > 2/3$ is used only in (P2) and (P3). $\square$

Remark 2.3 (how (P1) and (P3) are used). By (P1) and monotonicity, every number of the form $2^a(T+1)^b(N+16)^c$ with $a \leq 80$ and $b, c \leq 12$ is at most $R^{\varepsilon/8}$; we use this without comment, saying “by (P1)”. With (P4) (for $I_* \leq T$) it gives

$$2^7 T^2 (N^2 + 1) \leq R^{\varepsilon/8}, \quad 160(N+4) \leq R^{\varepsilon/8}, \quad 16 I_* \leq R^{\varepsilon/8}. \quad (2.2)$$

It also gives $R^{\varepsilon/8} \geq 2^{80}$, so every absolute constant $C \leq 2^{90}$ satisfies $C \leq R^{\varepsilon}$. Since $\varepsilon T \geq 240$,

$$R^{-3\varepsilon T/8} \leq R^{-90}. \quad (2.3)$$

By (P3), $(N+14)^{7/\kappa+1/\varepsilon+1} \leq R^{\kappa/4}$, and $C^{1/\varepsilon} \leq R^{\kappa/4}$ for every absolute constant $C \leq 2^{10}$.

Remark 2.4 (orders of magnitude). At scale N the gain is $R^\beta = 2^{c_\theta N^\theta}$. The losses that must stay below it are $R^{C\kappa}$ (block boundaries and short intervals), $R^{C\varepsilon/\kappa}$ (one factor $R^{C\varepsilon}$ for each of $O(1/\kappa)$ steps), and $2^{O(1/\varepsilon)}$ or $N^{O(1/\kappa+1/\varepsilon)}$ (the number of regular parts, the constants of the $O(1/\varepsilon)$ low-frequency steps, and the number of chains). The choices $\kappa \approx \beta/1000$ and $\varepsilon \approx \kappa^2/10^6$ keep the first two at most $R^{0.6\beta}$, and (P2), (P3) keep the last ones at most R^κ . The last requirement reads $\varepsilon^{-1} \log N \lesssim \kappa N$, that is $\kappa^3 N \gg \log N$, which is where $\theta > 2/3$ comes from.

3 Analytic tools with explicit constants

Since “negligible” means at most R^{-20} , and each integration by parts gains only about $R^{-\varepsilon}$, we need about $1/\varepsilon \asymp N^{2-2\theta}$ integrations by parts. So all derivative bounds must be tracked up to order $O(T)$, with explicit dependence on the order. This section collects the tools.

Definition 3.1 (regular functions). Let $A \geq 0$, $M \geq 1$ and $k_0 \in \mathbb{N}$. A complex function f on an interval or on the circle $\mathbb{R}/2\pi\mathbb{Z}$ is (A, M) -regular up to order k_0 if it is C^{k_0} and $|f^{(k)}| \leq AM^k$ for $0 \leq k \leq k_0$.

By the Leibniz rule, if f is (A, M) -regular and $\tilde{f}$ is $(\tilde{A}, \tilde{M})$ -regular up to order k_0 , then $f\tilde{f}$ is $(A\tilde{A}, M + \tilde{M})$ -regular up to order k_0 , because $\sum_i \binom{k}{i} M^i \tilde{M}^{k-i} = (M + \tilde{M})^k$.

3.1 A bump function with explicit derivatives

Lemma 3.2 (bump). *Let $a_j = 6/(\pi^2 j^2)$ for $j \geq 1$, so that $\sum_j a_j = 1$. There is a smooth even probability density ω on $\mathbb{R}$, supported in $[-1, 1]$, with*

$$\hat{\omega}(\xi) = \prod_{j \geq 1} \frac{\sin(a_j \xi)}{a_j \xi} \quad \text{and} \quad \|\omega^{(k)}\|_{L^1} \leq \left(\frac{\pi^2}{6}\right)^k (k!)^2 \quad (k \geq 0).$$

Proof. Let $\Omega(\xi) = \prod_{j \geq 1} \text{sinc}(a_j \xi)$, $\text{sinc}(y) = \sin y/y$. Since $|1 - \text{sinc}(y)| \leq y^2/6$ and $\sum a_j^2 < \infty$, the product converges uniformly on compact sets, and it is the limit of the Fourier transforms of the probability densities $\varpi_n = *_{j \leq n} (2a_j)^{-1} \mathbf{1}_{[-a_j, a_j]}$, which are supported in $[-1, 1]$. Since $|\text{sinc}(y)| \leq \min(1, 1/|y|)$, for every k we have $|\Omega(\xi)| \leq \prod_{j \leq k+2} \min(1, (a_j |\xi|)^{-1})$, so $(1+|\xi|)^k \Omega \in L^1$ for all k . Hence ω , the inverse Fourier transform of Ω , is C^∞ . By Lévy's continuity theorem $\varpi_n \rightarrow \omega$ weakly, so $\omega \geq 0$, $\int \omega = 1$ and $\text{supp } \omega \subseteq [-1, 1]$; ω is even because Ω is real and even.

Fix k . The same argument shows that $\Omega_k = \prod_{j > k} \text{sinc}(a_j \cdot)$ is the Fourier transform of a probability density ψ_k . The Fourier transform of the signed measure $(\delta_{-a} - \delta_a)/(2a)$ is $i \sin(a\xi)/a = i\xi \text{sinc}(a\xi)$. Hence

$$(i\xi)^k \Omega(\xi) = \prod_{j \leq k} \frac{i \sin(a_j \xi)}{a_j} \cdot \Omega_k(\xi), \quad \text{that is,} \quad \omega^{(k)} = \nu_k * \psi_k, \quad \nu_k = *_{j \leq k} \frac{\delta_{-a_j} - \delta_{a_j}}{2a_j}.$$

The total variation of ν_k is at most $\prod_{j \leq k} a_j^{-1} = (\pi^2/6)^k (k!)^2$, and $\|\psi_k\|_{L^1} = 1$. $\square$

For $0 < \delta \leq 1$ put $\omega_\delta(t) = \delta^{-1} \omega(t/\delta)$. On the circle $\mathbb{R}/2\pi\mathbb{Z}$ we convolve periodically; since $\delta \leq 1 < \pi$ this is the same as convolving on $\mathbb{R}$ locally.

Corollary 3.3 (cutoffs). *Let $0 < \delta \leq 1$, let E be a measurable subset of $\mathbb{R}$ or of $\mathbb{R}/2\pi\mathbb{Z}$, and let E_δ denote its closed δ -neighbourhood. Then $f = \mathbf{1}_{E_\delta} * \omega_\delta$ satisfies $0 \leq f \leq 1$, $f = 1$ on E , $f = 0$ outside $E_{2\delta}$, and $|f^{(k)}| \leq (\pi^2/6)^k (k!)^2 \delta^{-k}$; in particular f is $(1, 2k_0^2/\delta)$ -regular up to order k_0 , for every $k_0 \geq 1$. If (E_i) is a finite or countable measurable partition of $\mathbb{R}/2\pi\mathbb{Z}$, the functions $f_i = \mathbf{1}_{E_i} * \omega_\delta$ satisfy $\sum_i f_i = 1$, vanish outside $(E_i)_\delta$, and obey the same derivative bounds.*

Proof. $f^{(k)} = \mathbf{1}_{E_\delta} * \omega_\delta^{(k)}$ and $\|\omega_\delta^{(k)}\|_{L^1} = \delta^{-k} \|\omega^{(k)}\|_{L^1}$. If $t \in E$ then $t - s \in E_\delta$ for $|s| \leq \delta$, so $f(t) = 1$; if $\text{dist}(t, E) > 2\delta$ then $t - s \notin E_\delta$ for $|s| \leq \delta$, so $f(t) = 0$. For $k \leq k_0$, $(\pi^2/6)^k (k!)^2 \leq (2k_0^2)^k$. The statements about f_i are proved in the same way. $\square$

We fix three cutoffs once and for all:

- $\chi_0 = \mathbf{1}_{[-\pi/4, \pi/4]} * \omega_{\pi/12}$ on $\mathbb{R}$: it equals 1 on $[-\pi/6, \pi/6]$, vanishes outside $[-\pi/3, \pi/3]$, and $|\chi_0^{(k)}| \leq (2\pi)^k (k!)^2$.
- $\Psi = \mathbf{1}_{[3/8, 3]} * \omega_{1/8}$ on $\mathbb{R}$: it equals 1 on $[1/2, 2]$, vanishes outside $[1/4, 4]$, and $|\Psi^{(k)}| \leq 14^k (k!)^2$.
- For an arc $\Gamma \subset S^1$ of length at most $1/10$: $\psi_\Gamma = \mathbf{1}_{\Gamma_{0.15}} * \omega_{0.05}$. It equals 1 on $\Gamma_{0.1}$, vanishes outside $\Gamma_{0.2}$, hence vanishes on the $(\pi/3)$ -neighbourhood of the opposite arc $-\Gamma$, and $|\psi_\Gamma^{(k)}| \leq 33^k (k!)^2$.

Averages over frequency. For a function f of $r > 0$ and a depth v ,

$$\text{avg}_{r \approx 2^v} f(r) = 2^{-v} \int_0^\infty \Psi(r/2^v) f(r) dr.$$

Since $\Psi \geq \mathbf{1}_{[1/2, 2]}$, for $f \geq 0$ we have $\int_{2^{v-1}}^{2^{v+1}} f(r) r^{-1} dr \leq 2 \text{avg}_{r \approx 2^v} f$. Also $\text{avg}_{r \approx 2^v} r^2 \leq 64 \cdot 4^v$.

3.2 Non-stationary phase

Lemma 3.4 (non-stationary phase). *Let $\mathbb{I}$ be a compact interval or the circle $\mathbb{R}/2\pi\mathbb{Z}$, and let $k_0 \geq 1$. Let a be (A, M) -regular up to order k_0 on $\mathbb{I}$, compactly supported in the interior of $\mathbb{I}$ if $\mathbb{I}$ is an interval. Let F be a real function, C^{k_0+1} on an open set $U \supseteq \text{supp } a$, with $|F'| \geq \lambda > 0$ and $|F^{(j)}| \leq \Lambda_F$ for $2 \leq j \leq k_0 + 1$ on U . Then for $1 \leq k \leq k_0$,*

$$\left| \int_{\mathbb{I}} e^{iF} a \right| \leq |\text{supp } a| \cdot A \cdot \left(\frac{2(k+1)^2 \max(M, \Lambda_F/\lambda)}{\lambda} \right)^k.$$

Linear phases. If F' is constant with $|F'| \geq \lambda$, and $a \in C^k$ (supported as above) satisfies $|a^{(k)}| \leq A'M'^k$ for some $A' \geq 0$ and some $M' > 0$ (possibly $M' < 1$), then $|\int_{\mathbb{I}} e^{iF} a| \leq |\text{supp } a| \cdot A' \cdot (M'/\lambda)^k$.

Proof. Let $u = 1/F'$ on U . Write $u = G \circ F'$ with $G(y) = 1/y$, so $|G^{(j)}(F')| = j!|F'|^{-j-1} \leq j!\lambda^{-j-1}$. By Faà di Bruno's formula and $|F^{(j)}| \leq \Lambda_F$,

$$|u^{(m)}| \leq \sum_{j=1}^m j! \lambda^{-j-1} S(m, j) \Lambda_F^j \leq \lambda^{-1} \max(1, \Lambda_F/\lambda)^m a(m), \quad a(m) = \sum_j j! S(m, j),$$

where $S(m, j)$ are Stirling numbers of the second kind (we used $B_{m,j}(1, 1, \dots) = S(m, j)$). The ordered Bell numbers satisfy $a(m) = \sum_{j=1}^m \binom{m}{j} a(m-j)$, $a(0) = 1$, and by induction $a(m) \leq m! (3/2)^m$, since $\sum_{j \geq 1} (2/3)^j / j! = e^{2/3} - 1 < 1$. So $|u^{(m)}| \leq \lambda^{-1} m! (\frac{3}{2} \max(1, \Lambda_F/\lambda))^m$ for $m \leq k_0$.

For a C^1 function f supported in $\text{supp } a$, integration by parts gives $\int e^{iF} f = \int (e^{iF})' \frac{f}{iF'} = i \int e^{iF} (uf)'$, with no boundary terms. Hence $\int e^{iF} a = i^k \int e^{iF} D^k a$, where $Df = (uf)'$. Expanding, $D^k a$ is a sum of at most $(k+1)!$ products $u^{(m_1)} \dots u^{(m_k)} a^{(m_0)}$ with $m_0 + \dots + m_k = k$; indeed each application of D multiplies a product of s factors by u and differentiates it, producing $s+1$ terms. Each product is at most

$$\lambda^{-k} \left(\prod_{l \geq 1} m_l! \right) \left(\frac{3}{2} \max(1, \Lambda_F/\lambda) \right)^{k-m_0} A M^{m_0} \leq A \lambda^{-k} k! \left(2 \max(M, \Lambda_F/\lambda) \right)^k.$$

Since $(k+1)! k! \leq (k+1)^{2k}$, we get $|D^k a| \leq A (2(k+1)^2 \max(M, \Lambda_F/\lambda)/\lambda)^k$, and $D^k a$ vanishes outside $\text{supp } a$. If F' is constant, then $u = 1/F'$ is constant, $|u| \leq 1/\lambda$ and $D^k a = u^k a^{(k)}$, which gives the statement about linear phases. $\square$

3.3 Stationary phase on the circle

Lemma 3.5 (quadratic phase). *Let $T \geq 1$, let $g \in C_c^{2T+2}(\mathbb{R})$, and let $\Lambda > 0$. Then*

$$\int_{\mathbb{R}} e^{i\Lambda s^2/2} g(s) ds = \left(\frac{2\pi}{\Lambda} \right)^{1/2} e^{i\pi/4} \sum_{j < T} \left(\frac{i}{2\Lambda} \right)^j \frac{g^{(2j)}(0)}{j!} + E,$$

where

$$|E| \leq \left(\frac{2}{\Lambda}\right)^{1/2} \frac{(2\Lambda)^{-T}}{T!} \left(\|g^{(2T)}\|_{L^1} + \|g^{(2T+2)}\|_{L^1} \right).$$

Proof. Since $g \in C_c^{2T+2}$, $|\hat{g}(\xi)| \leq C(1+|\xi|)^{-2T-2}$, so $\xi^{2j}\hat{g}$ is integrable for $j \leq T$. The Fourier transform of the tempered function $e^{i\Lambda s^2/2}$ is $(2\pi/\Lambda)^{1/2}e^{i\pi/4}e^{-i\xi^2/(2\Lambda)}$, so by Parseval (valid for such g by mollification, since $\hat{g} \in L^1$; the phase is even)

$$\int e^{i\Lambda s^2/2} g(s) ds = \left(\frac{2\pi}{\Lambda}\right)^{1/2} e^{i\pi/4} \frac{1}{2\pi} \int e^{-i\xi^2/(2\Lambda)} \hat{g}(\xi) d\xi.$$

For real y , $|e^{-iy} - \sum_{j < T} (-iy)^j/j!| \leq |y|^T/T!$. Since $\frac{1}{2\pi} \int \xi^{2j} \hat{g}(\xi) d\xi = (-1)^j g^{(2j)}(0)$, the terms $j < T$ give the sum. For the remainder, $|\xi|^{2T} |\hat{g}(\xi)| \leq \min(\|g^{(2T)}\|_{L^1}, \xi^{-2} \|g^{(2T+2)}\|_{L^1})$, and $\int \min(a, b\xi^{-2}) d\xi = 4\sqrt{ab} \leq 2(a+b)$. Finally $(2\pi)^{1/2} \cdot \frac{1}{2\pi} \cdot 2 \leq 2^{1/2}$. $\square$

Lemma 3.6 (the stationary phase operators). *Let $\vartheta(s) = 2 \arcsin(s/2)$ for $|s| < 2$. There are real numbers p'_{jk} ($j \geq 0$, $0 \leq k \leq 2j$) with $p'_{00} = 1$ and $|p'_{jk}| \leq 1.2(2j)!(3.7)^k$, such that for every ϕ_0 and every G of class C^{2j} near ϕ_0 ,*

$$\partial_s^{2j} [G(\phi_0 + \vartheta(s)) \vartheta'(s)]_{s=0} = \sum_{k \leq 2j} p'_{jk} G^{(k)}(\phi_0).$$

Define the constant-coefficient operators

$$P_j = \frac{(i/2)^j}{j!} \sum_k p'_{jk} \partial^k, \quad \bar{P}_j = \frac{(-i/2)^j}{j!} \sum_k p'_{jk} \partial^k.$$

For a polynomial $P(\partial) = \sum_k c_k \partial^k$ put $\|P\|_M = \sum_k |c_k| M^k$. Then $P_0 = \bar{P}_0 = 1$ and, for $j \geq 1$ and $M \geq 1$,

$$\|P_j\|_M = \|\bar{P}_j\|_M \leq (100jM^2)^j.$$

Proof. By Faà di Bruno's formula the left side is a linear combination of $G^{(k)}(\phi_0)$, $k \leq 2j$, whose coefficients are polynomials in the real numbers $\vartheta^{(m)}(0)$. Taking $G(\phi) = e^{z(\phi - \phi_0)}$ shows that $\sum_k p'_{jk} z^k = \partial_s^{2j} [e^{z\vartheta(s)} \vartheta'(s)]_{s=0}$. The function $s \mapsto e^{z\vartheta(s)} \vartheta'(s)$ is holomorphic in $|s| < 2$. On $|s| = 1$: $\arcsin t = \sum_l c_l t^{2l+1}$ with $0 < c_l \leq 1$, so $|\vartheta(s)| \leq 2 \cdot \frac{1/2}{1-1/4} = \frac{4}{3}$; and $\vartheta'(s) = (1 - s^2/4)^{-1/2}$ with $|1 - s^2/4| \geq \frac{3}{4}$, so $|\vartheta'(s)| \leq 1.16$. By Cauchy's estimate, $|\sum_k p'_{jk} z^k| \leq (2j)! 1.16 e^{4|z|/3}$, and by Cauchy's estimate on $|z| = \rho$, $|p'_{jk}| \leq 1.16 (2j)! e^{4\rho/3} \rho^{-k}$. Take $\rho = 3k/4$ for $k \geq 1$, which gives $(4e/(3k))^k \leq (3.7)^k$, and let $\rho \rightarrow 0$ for $k = 0$. Also $p'_{00} = \vartheta'(0) = 1$. Finally, using $(2j)!/j! \leq (2j)^j$,

$$\|P_j\|_M \leq \frac{1.16(2j)!}{2^j j!} \sum_{k \leq 2j} (3.7M)^k \leq 1.16(2j+1)(2j)^j 2^{-j} (3.7M)^{2j} \leq (100jM^2)^j,$$

because $1.16(2j+1)(13.7)^j \leq 100^j$ for $j \geq 1$. $\square$

Lemma 3.7 (stationary phase on the circle). *Let $\Lambda \geq 1$, $T \geq 1$, $\phi_0 \in \mathbb{R}$, and let G be a function on $\mathbb{R}/2\pi\mathbb{Z}$ that is (A, M) -regular up to order $2T+2$, with $M \geq 1$. Then*

$$\begin{aligned} & \int_0^{2\pi} e^{-i\Lambda \cos(\phi - \phi_0)} G(\phi) d\phi \\ &= \left(\frac{2\pi}{\Lambda}\right)^{1/2} \left[e^{-i(\Lambda - \pi/4)} \sum_{j < T} \Lambda^{-j} (P_j G)(\phi_0) + e^{i(\Lambda - \pi/4)} \sum_{j < T} \Lambda^{-j} (\bar{P}_j G)(\phi_0 + \pi) \right] + E, \end{aligned}$$

with

$$|E| \leq A (10^8(T+1)^4 M^2)^{T+1} \Lambda^{-T}.$$

Proof. Split $G = \chi_0(\phi - \phi_0)G + \chi_0(\phi - \phi_0 - \pi)G + cG$, where the cutoffs are periodized (well defined since $\text{supp } \chi_0 \subseteq [-\pi/3, \pi/3]$) and $c = 1 - \chi_0(\cdot - \phi_0) - \chi_0(\cdot - \phi_0 - \pi)$; the two cutoffs have disjoint supports, so $0 \leq c \leq 1$.

Near ϕ_0 . Substitute $\phi = \phi_0 + \vartheta(s)$, $|s| \leq 1$; then $\cos(\phi - \phi_0) = 1 - s^2/2$ and $d\phi = \vartheta'(s) ds$. The integral becomes $e^{-i\Lambda} \int e^{i\Lambda s^2/2} g(s) ds$ with $g = (\chi_0 \circ \vartheta) \cdot (G \circ (\phi_0 + \vartheta)) \cdot \vartheta'$. This g is C_c^{2T+2} on $(-2, 2)$ and vanishes for $1 \leq |s| < 2$, where $|\vartheta(s)| \geq \pi/3$; so its extension by zero lies in $C_c^{2T+2}(\mathbb{R})$. Since $\chi_0 \circ \vartheta = 1$ near 0, Lemma 3.6 gives $g^{(2j)}(0) = \sum_k p'_{jk} G^{(k)}(\phi_0)$, and Lemma 3.5 produces the first sum, with an error E_1 .

To bound E_1 we need derivatives of g on $[-1, 1]$. For real $|s| \leq 1$, ϑ is holomorphic on the disc of radius $1/2$ about s , where $|\vartheta| < 3.5$ and $|\vartheta'| < 1.52$; by Cauchy's estimate $|\vartheta^{(m)}(s)| \leq 3.5 m! 2^m$ and $|(\vartheta')^{(m)}(s)| \leq 1.52 m! 2^m$. By Faà di Bruno's formula, with $B_{n,k}(1!, 2!, \dots) = \binom{n-1}{k-1} \frac{n!}{k!}$ (Lah numbers),

$$|(G \circ (\phi_0 + \vartheta))^{(n)}| \leq \sum_{k=1}^n A M^k 3.5^k 2^n \binom{n-1}{k-1} \frac{n!}{k!} \leq A n! (14M)^n, \quad |(\chi_0 \circ \vartheta)^{(n)}| \leq (n!)^2 88^n,$$

using $|\chi_0^{(k)}| \leq (2\pi)^k (k!)^2$ for the second. By the Leibniz rule for three factors,

$$|g^{(n)}| \leq 1.52 A (n!)^2 (88M)^n \binom{n+2}{2} \leq 1.52 A (n!)^2 (264M)^n, \quad \text{so} \quad \|g^{(n)}\|_{L^1} \leq 4A(n!)^2 (264M)^n.$$

Using $((2T+2)!)/T! \leq (2T+2)^4 (87(T+1)^3)^T$, Lemma 3.5 gives

$$\begin{aligned} |E_1| &\leq \left(\frac{2}{\Lambda}\right)^{1/2} \frac{(2\Lambda)^{-T}}{T!} 8A ((2T+2)!)^2 (264M)^{2T+2} \\ &\leq 1.3 \cdot 10^7 A (T+1)^4 M^2 (3.1 \cdot 10^6 (T+1)^3 M^2)^T \Lambda^{-T}. \end{aligned}$$

Near $\phi_0 + \pi$. Substitute $\phi = \phi_0 + \pi + \vartheta(s)$; now $\cos(\phi - \phi_0) = -1 + s^2/2$, and the integral is $e^{i\Lambda} \int e^{-i\Lambda s^2/2} \tilde{g}(s) ds$. Applying Lemma 3.5 to $\tilde{g}$ and conjugating gives the second sum (with $(-i/(2\Lambda))^j$, hence $\bar{P}_j$) and an error E_2 with the same bound.

Away from both. On $\text{supp } c$ we have $|\sin(\phi - \phi_0)| \geq \frac{1}{2}$, and $|\sin(\phi - \phi_0)| > 0.49$ on an open neighbourhood of $\text{supp } c$. Put $F = -\Lambda \cos(\phi - \phi_0)$, so $|F'| \geq \Lambda/3$ there and $|F^{(j)}| \leq \Lambda$. The function c is $(2, 2\pi T^2)$ -regular up to order T (since $(k!)^2 \leq T^{2k}$ for $k \leq T$), so cG is $(2A, 8T^2 M)$ -regular up to order T . Lemma 3.4 with $k = T$, $\lambda = \Lambda/3$, $\Lambda_F = \Lambda$ gives

$$\left| \int e^{iF} cG \right| \leq 2\pi \cdot 2A \left(\frac{2(T+1)^2 \max(8T^2 M, 3)}{\Lambda/3} \right)^T \leq 4\pi A (48(T+1)^4 M)^T \Lambda^{-T}.$$

Adding the three errors gives the bound on E . □

Lemma 3.8 (inverting the expansion). *Define $Q_0 = 1$ and $Q_j = -\sum_{i=1}^j P_i Q_{j-i}$ for $j \geq 1$ (products of polynomials in ∂). Then $\sum_{i+j=m} P_i Q_j = 0$ for every $m \geq 1$, and for $M \geq 1$:*

$$\|Q_j\|_M \leq (200 j M^2)^j \quad (j \geq 1), \quad \left\| \sum_{\substack{i+j=m \\ i,j < T}} P_i Q_j \right\|_M \leq (m+1)(200 m M^2)^m \quad (m \geq 1).$$

Proof. The identity holds by construction since $P_0 = 1$. The norm $\|\cdot\|_M$ is submultiplicative. By induction, using $i^i(j-i)^{j-i} \leq j^j$,

$$\|Q_j\|_M \leq \sum_{i=1}^j (100 i M^2)^i (200(j-i)M^2)^{j-i} \leq (jM^2)^j \sum_{i=1}^j 100^i 200^{j-i} \leq (200 j M^2)^j.$$

The second bound follows in the same way. $\square$

3.4 Separation of variables

Lemma 3.9 (separation of variables). *Let $X \subseteq B(x_0, \varrho)$ and $Y \subseteq B(y_0, \varrho)$ with $D = |x_0 - y_0| \geq 50\varrho$. For every integer $j \geq 1$ there are polynomials f_k, g_k ($k \in \mathbb{N}$) with $|f_k| \leq 1$ on X and $|g_k| \leq 1$ on Y , and real numbers c_k with $\sum_k |c_k| \leq 2(\sqrt{2}/D)^j$, such that*

$$|x - y|^{-j} = \sum_k c_k f_k(x) g_k(y) \quad (x \in X, y \in Y),$$

the series converging absolutely and uniformly.

Proof. Write $x = x_0 + \varrho u$, $y = y_0 + \varrho u'$ with $|u|, |u'| \leq 1$, $\hat{c} = (x_0 - y_0)/D$ and $w = (\varrho/D)(u - u')$. Then $|x - y| = D|\hat{c} + w|$ and $|\hat{c} + w|^{-j} = (1 + s)^{-j/2}$ with $s = 2\hat{c} \cdot w + |w|^2$. As a polynomial in the four coordinates of u, u' , s has absolute coefficient sum at most $4\sqrt{2}/50 + 8/2500 < 0.12$, and every monomial is at most 1 in absolute value. Expand $(1 + s)^{-j/2} = \sum_m \binom{-j/2}{m} s^m$, with $|\binom{-j/2}{m}| \leq 2^{j/2+m}$, and expand each s^m into monomials. Each monomial is a product of a monomial in u and a monomial in u' . The absolute coefficient sum is at most $2^{j/2} \sum_m (0.24)^m \leq 2 \cdot 2^{j/2}$. $\square$

4 Distance measures and a summable criterion

Definition 4.1 (distance measure). Let μ_1, μ_2 be finite positive measures on $\mathbb{R}^2$ whose supports are disjoint compact sets. Their *distance measure* is the measure ν on $(0, \infty)$ given by

$$\int f d\nu = \iint f(|x - y|) |x - y|^{-1/2} d\mu_1(x) d\mu_2(y).$$

The weight $|x - y|^{-1/2}$ is bounded above and below on the supports. The measure ν is carried by $\{|x - y| : x \in \text{supp } \mu_1, y \in \text{supp } \mu_2\}$, so: *if μ_1, μ_2 live on E and their distance measure is absolutely continuous and nonzero, then $|\Delta(E)| > 0$.*

The following criterion is [9, Lemma 7]. It replaces the power-decay criterion of [7, Lemma 2.2]: the removed mass only has to be summable.

Lemma 4.2 (summable criterion). *Let α be a finite positive measure on $\mathbb{R}$. Suppose that for all $N \geq N_0$ there is a positive measure $\tau_N \leq \alpha$ such that*

$$\sum_{N \geq N_0} (\alpha(\mathbb{R}) - \tau_N(\mathbb{R})) < \infty \quad \text{and} \quad \sum_{N \geq N_0} \int_{2^{N-1} \leq |r| \leq 2^{N+1}} |\hat{\tau}_N(r)|^2 dr < \infty.$$

Then α is absolutely continuous.

Proof. Fix a smooth even $\varphi : \mathbb{R} \rightarrow [0, 1]$ with $\varphi = 1$ on $[-1, 1]$ and $\varphi = 0$ outside $[-2, 2]$. For a finite measure ν let $Q_M \nu$ be the function with Fourier transform $\varphi(2^{-M}r)\hat{\nu}(r)$, that is, ν convolved with $2^M \check{\varphi}(2^M \cdot)$, and put $P_M = Q_M - Q_{M-1}$. The multiplier $m_M(r) = \varphi(2^{-M}r) - \varphi(2^{1-M}r)$ of P_M satisfies $|m_M| \leq 1$ and vanishes outside $2^{M-1} \leq |r| \leq 2^{M+1}$; at each r at most three of the m_M are nonzero. Hence

$$\|P_M \nu\|_{L^1} \leq 2\|\check{\varphi}\|_{L^1}\|\nu\|, \quad \left\| \sum_{M=M_1}^{M_2} P_M \tau_M \right\|_{L^2}^2 \leq \frac{3}{2\pi} \sum_{M=M_1}^{M_2} \int_{2^{M-1} \leq |r| \leq 2^{M+1}} |\hat{\tau}_M(r)|^2 dr.$$

For $M > N_0$ write $P_M \alpha = P_M \tau_M + P_M(\alpha - \tau_M)$. By the hypotheses, $\sum_M P_M \tau_M$ converges in L^2 , and $\sum_M P_M(\alpha - \tau_M)$ converges in L^1 , because $\|\alpha - \tau_M\| = \alpha(\mathbb{R}) - \tau_M(\mathbb{R})$. Since $Q_M \alpha = Q_{N_0} \alpha + \sum_{N_0 < m \leq M} P_m \alpha$ tends to α in the sense of distributions and $Q_{N_0} \alpha \in L^1$, the measure α equals the sum of an L^1 function and an L^2 function. A finite measure given by a locally integrable function is absolutely continuous. $\square$

5 Preparation

5.1 Frostman's lemma for the gauge

Lemma 5.1 (Frostman). *Let $E \subset \mathbb{R}^2$ be compact with $\mathcal{H}^{h_\theta}(E) > 0$. There is a Borel probability measure ζ with $\text{supp } \zeta \subseteq E$ and a constant C_E such that $\zeta(B(x, r)) \leq C_E h_\theta(r)$ for all $x \in \mathbb{R}^2$ and $0 < r < 1$.*

Proof. This is the usual dyadic construction [6, Theorem 8.8]; we record it because the gauge is not a power. Write $h = h_\theta$, and recall $h(2r) \leq 4h(r)$ for $r \leq 1/(2e)$ by (1.1). Let $\mathcal{D}_k$ be the half-open dyadic squares of side 2^{-k} , and fix k_0 with $2^{3-k_0} \leq 1/(2e)$. For $m \geq k_0$ let $\mu_m^{(m)}$ give each $Q \in \mathcal{D}_m$ that meets E the mass $h(2^{-m})$, spread uniformly on Q . For $k = m-1, \dots, k_0$, obtain $\mu_m^{(k)}$ from $\mu_m^{(k+1)}$ by multiplying it, on each $Q \in \mathcal{D}_k$ with $\mu_m^{(k+1)}(Q) > h(2^{-k})$, by $h(2^{-k})/\mu_m^{(k+1)}(Q)$. Put $\mu_m = \mu_m^{(k_0)}$. Then $\mu_m(Q) \leq h(2^{-k})$ for all $Q \in \mathcal{D}_k$, $k_0 \leq k \leq m$. Moreover every $y \in E$ lies in some $Q \in \mathcal{D}_k$, $k_0 \leq k \leq m$, with $\mu_m(Q) = h(2^{-k})$: take the coarsest ancestor of the m -square of y at which a reduction occurred, or the m -square itself if there was none. The maximal such squares Q_i are disjoint and cover E , and $h(\text{diam } Q_i) \leq h(2\ell(Q_i)) \leq 4h(\ell(Q_i))$, so

$$\mu_m(\mathbb{R}^2) \geq \sum_i h(\ell(Q_i)) \geq \frac{1}{4} \mathcal{H}_{\delta_0}^h(E) =: c_E, \quad \delta_0 = \sqrt{2} 2^{-k_0}.$$

Here $c_E > 0$: if $\mathcal{H}_{\delta_0}^h(E) = 0$, then E has covers by sets of diameter less than 1 with $\sum h(\text{diam } U_i)$ arbitrarily small, hence with all diameters arbitrarily small (as h is increasing and positive on $(0, 1)$), so $\mathcal{H}^{h_\theta}(E) = 0$. Let ζ be a weak limit of $\mu_m/\mu_m(\mathbb{R}^2)$; it is a probability measure carried by E . For $x \in \mathbb{R}^2$ and $r \leq 2^{-k_0-3}$ pick $k \geq k_0$ with $4r \leq 2^{-k} < 8r$. Each coordinate projection of the open disc $B^\circ(x, 2r)$ is an open interval of length $4r \leq 2^{-k}$, which meets at most two half-open dyadic intervals of length 2^{-k} ; so the disc meets at most four squares of $\mathcal{D}_k$, and

$$\zeta(B(x, r)) \leq \zeta(B^\circ(x, 2r)) \leq \liminf_m \frac{\mu_m(B^\circ(x, 2r))}{\mu_m(\mathbb{R}^2)} \leq \frac{4h(2^{-k})}{c_E} \leq \frac{4h(8r)}{c_E} \leq \frac{256 h(r)}{c_E}.$$

For $2^{-k_0-3} < r < 1$ use $\zeta(B(x, r)) \leq 1 \leq h(r)/h(2^{-k_0-3})$. $\square$

5.2 Orlicz estimates for radial projections

This subsection follows [9, §§1–3], with the details written out. For $\gamma \geq 1$ put

$$\Phi_\gamma(t) = t [\log(e+t)]^\gamma \quad (t \geq 0), \quad L(r) = 1 + \log_+(1/r), \quad J_\gamma(\rho) = \iint \frac{L(|u-v|)^\gamma}{|u-v|} d\rho(u) d\rho(v),$$

$$\mathcal{E}_\gamma(\rho) = \int_{\mathbb{R}^2} |\hat{\rho}(\xi)|^2 \frac{[\log(e+|\xi|)]^\gamma}{|\xi|} d\xi.$$

Φ_γ is increasing, convex and superlinear. For $x \in \mathbb{R}^2$ let $\pi_x(y) = (y-x)/|y-x|$ (so $\pi_x(y) = e(y, x)$), and for $w \in S^1$ let $P_w(x) = x \cdot w$.

Lemma 5.2 (Fourier energy). *For every Borel probability measure ρ on $\mathbb{R}^2$, $\mathcal{E}_\gamma(\rho) \leq C_\gamma(1 + J_\gamma(\rho))$.*

Proof. The region $|\xi| < 1$ contributes at most $2\pi[\log(e+1)]^\gamma$. On the annulus $2^n \leq |\xi| < 2^{n+1}$ ($n \geq 0$) we have $[\log(e+|\xi|)]^\gamma |\xi|^{-1} \leq (2(n+1))^\gamma 2^{-n}$ and $1 \leq e \cdot e^{-|\xi|^2/4^{n+1}}$. Since $\int e^{-|\xi|^2/s^2} |\hat{\rho}(\xi)|^2 d\xi = \pi s^2 \iint e^{-s^2|u-v|^2/4} d\rho d\rho$, the annuli contribute at most

$$4\pi e 2^\gamma \iint K(|u-v|) d\rho(u) d\rho(v), \quad K(r) = \sum_{n \geq 0} (n+1)^\gamma 2^n e^{-4^n r^2}.$$

For $r \geq 1$, $K(r) \leq C_\gamma$. For $r < 1$ let $n_0 = \lfloor \log_2(1/r) \rfloor$. The terms $n \leq n_0$ add up to at most $2(n_0+1)^\gamma 2^{n_0} \leq 2(1 + \log_2(1/r))^\gamma / r$. For $n = n_0 + m$ with $m \geq 1$ we have $e^{-4^n r^2} \leq e^{-4^{m-1}}$, $(n+1)^\gamma \leq (n_0+1)^\gamma (m+1)^\gamma$ and $2^n \leq 2^m / r$; these terms add up to at most $C_\gamma(n_0+1)^\gamma / r$. Since $1 + \log_2(1/r) \leq 1.45 L(r)$, $K(r) \leq C_\gamma L(r)^\gamma / r$ for $r < 1$. $\square$

Lemma 5.3 (orthogonal projections). *Let ρ be a Borel probability measure on $\mathbb{R}^2$ with $\mathcal{E}_\gamma(\rho) < \infty$. There is a Borel function $f : S^1 \times \mathbb{R} \rightarrow [0, \infty)$ such that $(P_w)_* \rho = f(w, \cdot) dt$ for almost every w , and*

$$\int_{S^1} \int_{\mathbb{R}} f(w, t)^2 [\log(e + f(w, t))]^\gamma dt dw \leq C_\gamma(1 + \mathcal{E}_\gamma(\rho)).$$

Proof. In polar coordinates, $\int_{S^1} \int_{\mathbb{R}} |\hat{\rho}(tw)|^2 dt dw = 2 \int |\hat{\rho}(\xi)|^2 |\xi|^{-1} d\xi \leq 2\mathcal{E}_\gamma(\rho)$. The Fourier transform of $(P_w)_* \rho$ is $t \mapsto \hat{\rho}(tw)$, so for almost every w this projection has a density in L^2 . The measure $\Theta = \int \delta_w \otimes (P_w)_* \rho dw$ on $S^1 \times \mathbb{R}$ is therefore absolutely continuous with respect to $dw \otimes dt$ (a null set has null fibres for almost every w). Let f be its Radon–Nikodym derivative; by uniqueness of disintegration, $f(w, \cdot)$ is a density of $(P_w)_* \rho$ for almost every w . Write $f_w = f(w, \cdot)$.

Fix a Schwartz function φ on $\mathbb{R}$ with $0 \leq \hat{\varphi} \leq 1$, $\hat{\varphi} = 1$ on $[-1, 1]$ and $\text{supp } \hat{\varphi} \subseteq [-2, 2]$; let $\varphi_j(t) = 2^j \varphi(2^j t)$ and $A_0 = \|\varphi\|_\infty$. Since $(P_w)_* \rho$ has mass 1, $|f_w * \varphi_j| \leq A_0 2^j$. Hence on $\{f_w > 2A_0 2^j\}$ we have $f_w \leq 2|f_w - f_w * \varphi_j|$, and by Plancherel

$$\iint_{\{f > 2A_0 2^j\}} f^2 \leq 4 \int_{S^1} \frac{1}{2\pi} \int_{|t| > 2^j} |\hat{\rho}(tw)|^2 dt dw = \frac{4}{\pi} \int_{|\xi| > 2^j} \frac{|\hat{\rho}(\xi)|^2}{|\xi|} d\xi.$$

If $f \leq 2A_0$ then $[\log(e+f)]^\gamma \leq C$. If $f > 2A_0$, let $m_f = \#\{j \geq 0 : 2A_0 2^j < f\} \geq 1$; then $\log(e+f) \leq C(1+m_f)$ and $\sum_{j < m_f} (j+1)^{\gamma-1} \geq m_f^\gamma / \gamma$. So $[\log(e+f)]^\gamma \leq C_\gamma(1 + \sum_{j \geq 0} (j+1)^{\gamma-1} \mathbf{1}_{\{f > 2A_0 2^j\}})$, and

$$\iint f^2 [\log(e+f)]^\gamma \leq C_\gamma \iint f^2 + C_\gamma \int \frac{|\hat{\rho}(\xi)|^2}{|\xi|} \sum_{j: 2^j < |\xi|} (j+1)^{\gamma-1} d\xi \leq C_\gamma(1 + \mathcal{E}_\gamma(\rho)),$$

using $\iint f^2 = \frac{1}{\pi} \int |\hat{\rho}|^2 |\xi|^{-1}$ and $\sum_{2^j < |\xi|} (j+1)^{\gamma-1} \leq C_\gamma [\log(e+|\xi|)]^\gamma$. $\square$

Theorem 5.4 (Orlicz radial projections [9, Theorem 1]). *Let $\gamma \geq 1$, and let μ, ν be Borel probability measures on $\mathbb{R}^2$ with compact supports at distance $s_0 > 0$. Put $D_0 = \max(1, s_0 + \sup\{|x - y| : x \in \text{supp } \nu, y \in \text{supp } \mu\})$. If $J_\gamma(\mu) + J_\gamma(\nu) < \infty$, there is a Borel function $G : \mathbb{R}^2 \times S^1 \rightarrow [0, \infty)$ such that $(\pi_x)_* \mu = G(x, \cdot) dw$ for ν -almost every x , and*

$$\iint_{S^1} \Phi_\gamma(G(x, w)) dw d\nu(x) \leq C(\gamma, D_0)(1 + J_\gamma(\mu) + J_\gamma(\nu)).$$

Proof. Smoothing and the weighted density. Let $0 < \delta < s_0/2$, let $\tilde{\omega}_\delta$ be a smooth radial probability density supported in $B(0, \delta)$, and $m = \mu * \tilde{\omega}_\delta$. Fix $x \in \text{supp } \nu$. In polar coordinates about x , the measure $|y - x|^{-1} m(y) dy$ has radial projection $u_x(w) dw$ with $u_x(w) = \int_0^\infty m(x + sw) ds$. This is at most $\int_{\mathbb{R}} m(x + sw) ds = f^m(w^\perp, x \cdot w^\perp)$, where $f^m(w', \cdot)$ is the (continuous) density of $(P_{w'})_*(m dy)$ and $w^\perp$ is w rotated by $\pi/2$.

Transfer. Let $v(w', \cdot)$ be the density of $(P_{w'})_* \nu$ given by Lemma 5.3. Since Φ_γ is increasing and dw is rotation invariant,

$$\iint \Phi_\gamma(u_x(w)) dw d\nu(x) \leq \int_{S^1} \int_{\mathbb{R}} \Phi_\gamma(f^m(w, \tau)) v(w, \tau) d\tau dw.$$

The elementary inequality $ab[\log(e + a)]^\gamma \leq a^2[\log(e + a)]^\gamma + b^2[\log(e + b)]^\gamma$ ($a, b \geq 0$; use the first term if $b \leq a$, the second otherwise), Lemma 5.3 for $m dy$ and for ν , $|\hat{m}| \leq |\hat{\mu}|$, and Lemma 5.2 bound the right side by $C_\gamma(1 + J_\gamma(\mu) + J_\gamma(\nu))$.

Removing the weight. The radial projection of $m dy$ from x has density $\tilde{u}_x(w) = \int_0^\infty m(x + sw) ds \leq D_0 u_x(w)$, and $\Phi_\gamma(D_0 t) \leq D_0(1 + \log D_0)^\gamma \Phi_\gamma(t)$. Hence $G_\delta(x, w) = \tilde{u}_x(w)$, which is continuous in (x, w) , satisfies $\int \Phi_\gamma(G_\delta) d(\nu \otimes dw) \leq K$, with $K = C(\gamma, D_0)(1 + J_\gamma(\mu) + J_\gamma(\nu))$ independent of δ .

Limit. The measures $\Theta_\delta = G_\delta \cdot (\nu \otimes dw)$ converge weakly to $\Theta = \int \delta_x \otimes (\pi_x)_* \mu d\nu(x)$ as $\delta \rightarrow 0$. Indeed, for continuous F on $\mathbb{R}^2 \times S^1$, $\int F d\Theta_\delta = \int [\int F(x, \pi_x(y)) d\nu(x)] m(y) dy$, the bracket is continuous in y on the $s_0/2$ -neighbourhood of $\text{supp } \mu$, and $m dy \rightarrow \mu$ weakly. Since Φ_γ is superlinear, $\{G_\delta\}$ is uniformly integrable, hence relatively weakly compact in $L^1(\nu \otimes dw)$ by the Dunford–Pettis theorem [1]. Let G be a weak limit of G_{δ_k} ; it is nonnegative almost everywhere, and we take a Borel representative $G \geq 0$. Then $\Theta = G \cdot (\nu \otimes dw)$. On the closed convex cone of nonnegative functions in L^1 , the functional $H \mapsto \int \Phi_\gamma(H)$ is convex and, by Fatou’s lemma, strongly lower semicontinuous; by Mazur’s lemma it is weakly lower semicontinuous, so $\int \Phi_\gamma(G) \leq K$. Finally both $\int \delta_x \otimes (\pi_x)_* \mu d\nu(x)$ and $\int \delta_x \otimes G(x, w) dw d\nu(x)$ disintegrate Θ over its first marginal ν , so $(\pi_x)_* \mu = G(x, \cdot) dw$ for ν -almost every x . $\square$

Corollary 5.5. *If ρ is a compactly supported Borel probability measure with $\rho(B(x, r)) \leq C_0 h_\theta(r)$ for all x and $0 < r < 1$, then $J_\gamma(\rho) \leq 1 + (1 + \gamma)C_0 \int_0^\infty (1 + t)^\gamma e^{-t^\theta} dt < \infty$ for every $\gamma \geq 1$.*

Proof. Let $k(r) = L(r)^\gamma/r$; it is decreasing, tends to 0 at infinity, and $-k'(r) \leq (1 + \gamma)L(r)^\gamma r^{-2}$ for $r < 1$, $-k'(r) = r^{-2}$ for $r > 1$. For each u , $\int k(|u - v|) d\rho(v) = \int_0^\infty (-k'(r))\rho(B(u, r)) dr$. Use the hypothesis for $r < 1$ and $\rho \leq 1$ for $r \geq 1$, and substitute $t = \log(1/r)$. $\square$

5.3 Two prepared pieces

Proposition 5.6 (two prepared pieces). *Let E be compact with $\mathcal{H}^{h_\theta}(E) > 0$. After replacing E by a similar copy, there are Borel probability measures μ_1, μ_2 on compact sets $S_1, S_2 \subseteq E \cap [0, 1]^2$, and constants $C_0, \mathcal{K}_0 < \infty$, such that:*

- (a) (gauge) $\mu_j(B(x, r)) \leq C_0 h_\theta(r)$ for all $x \in \mathbb{R}^2$ and $0 < r < 1$;
- (b) (separation) $S_1 \subseteq B(x_0, 1/400)$ and $S_2 \subseteq B(y_0, 1/400)$, where $|x_0 - y_0| = 1/4$ and both discs lie in $[0, 1]^2$. Consequently $0.24 \leq |x - y| \leq 0.26$ for $x \in S_1, y \in S_2$, and all directions $e(x, y)$ lie in an arc Γ of length at most $1/20$;
- (c) (spread directions) for every $x \in S_1$ the radial projection $(\pi_x)_* \mu_2$ has a density F_x with $\int_{S^1} \Phi_8(F_x) \leq \mathcal{K}_0$; and for every $y \in S_2$ the radial projection $(\pi_y)_* \mu_1$ has a density with the same bound.

Similar copies are harmless: $|\Delta(\lambda E + z)| = \lambda |\Delta(E)|$, and by (1.1) the hypothesis $\mathcal{H}^{h_\theta}(E) > 0$ is preserved.

Proof. Let ζ be the measure of Lemma 5.1. Pick $x'_0 \neq y'_0$ in $\text{supp } \zeta$, let $\varrho' = |x'_0 - y'_0|/100$, and restrict ζ to $E \cap B(x'_0, \varrho')$ and to $E \cap B(y'_0, \varrho')$; both restrictions have positive mass. Apply the similarity that maps x'_0, y'_0 to $x_0 = (\frac{1}{2}, \frac{1}{2}) - \frac{1}{8}\hat{e}$ and $y_0 = (\frac{1}{2}, \frac{1}{2}) + \frac{1}{8}\hat{e}$, where $\hat{e}$ is the direction of $y'_0 - x'_0$; it maps the two discs to $B(x_0, 1/400)$ and $B(y_0, 1/400)$. Normalize the images to probability measures μ'_1, μ'_2 . By (1.1) they satisfy (a) with some constant, and (b) holds for their supports. The angle between $x - y$ and $x_0 - y_0$ is at most $\arcsin(\frac{1/200}{1/4 - 1/200}) < 0.021$.

By Corollary 5.5, $J_8(\mu'_j) < \infty$, and the supports are at distance at least 0.24. Apply Theorem 5.4 with $\gamma = 8$ in both directions; let G_2 be the function it gives for the radial projections of μ'_2 from points x , and let A_1 be a Borel set of full μ'_1 -measure on which $(\pi_x)_* \mu'_2 = G_2(x, \cdot) dw$ (the exceptional set is contained in a Borel null set). By Chebyshev's inequality there is $K_1 < \infty$ such that the Borel set $W'_1 = \{x \in A_1 : \int \Phi_8(G_2(x, \cdot)) \leq K_1\}$ has $\mu'_1(W'_1) \geq 1/2$; define W'_2 symmetrically. Choose compact $W_j \subseteq W'_j$ with $\mu'_j(W_j) \geq 1/4$, and put $\mu_j = \mu'_j|_{W_j}/\mu'_j(W_j)$, $S_j = \text{supp } \mu_j \subseteq W_j$. Then (a) holds with C_0 four times the previous constant, and (b) still holds. For $x \in S_1$, the radial projection of μ_2 from x is at most $4G_2(x, \cdot) dw$, so it has a density $F_x \leq 4G_2(x, \cdot)$, and $\int \Phi_8(F_x) \leq 4(1 + \log 4)^8 K_1 =: \mathcal{K}_0$. The other half of (c) is the same. $\square$

The threshold N_0 . From now on $N_0 \geq N_{\text{par}}(\theta)$ is also so large that $\log_2 C_0 \leq \kappa N/2$ for $N \geq N_0$. This is possible because $\kappa N \rightarrow \infty$.

5.4 Regular pieces

Definition 5.7 (regular measure). Let ρ be a probability measure on $[0, 1]^2$, and for each depth n let $M(n)$ be the largest ρ -mass of an n -cell. The measure ρ is *regular* if at every depth n every occupied n -cell has mass at least $R^{-\varepsilon} M(n)$.

Always $M(0) = 1$ and $M(n+1) \leq M(n) \leq 4M(n+1)$.

Lemma 5.8 (regular decomposition). Assume (P2). Let μ be a probability measure on $[0, 1]^2$. There are disjoint sets $G_1, \dots, G_m$, each a union of N -cells of positive mass, such that $\mu(G_i) \geq R^{-\kappa/2}$ for each i , $\mu([0, 1]^2 \setminus \bigcup_i G_i) \leq 2R^{-\kappa/4}$, and each $\mu|_{G_i}/\mu(G_i)$ is regular.

Proof. Let $\Delta = \lfloor \varepsilon N/4 \rfloor$ and $w = \lfloor \varepsilon N/8 \rfloor$. Since $\varepsilon N \geq 16$, $\Delta \geq \varepsilon N/8$ and $\varepsilon N/16 \leq w \leq \varepsilon N/8$. Let $k = \lceil N/\Delta \rceil$, $n_i = i\Delta$ for $0 \leq i < k$ and $n_k = N$; consecutive n_i differ by at most Δ , and $k \leq 8/\varepsilon + 1$. For $0 < m \leq 1$ let $\text{cl}(m) = \lfloor \log_2(1/m)/w \rfloor$, so that $2^{-w(\text{cl}(m)+1)} < m \leq 2^{-w \text{cl}(m)}$.

Step 0. The N -cells of mass less than R^{-3} have total mass at most $4^N R^{-3} = R^{-1}$. Discard them, and call the remaining N -cells *heavy*. Every union of heavy cells considered below has mass in $[R^{-3}, 1]$, so its class lies in $\{0, 1, \dots, \lfloor 3N/w \rfloor\}$, a set of at most $48/\varepsilon + 1$ values.

Types. For a heavy cell Q and $0 \leq i \leq k$ let $D_i(Q)$ be the n_i -cell containing Q . Define $S_k(Q) = Q$, $\ell_k(Q) = \text{cl}(\mu(Q))$ and $\tau_k(Q) = (\ell_k(Q))$; and, for $i = k-1, \dots, 0$,

$$S_i(Q) = \bigcup \{Q' : Q' \text{ heavy, } Q' \subseteq D_i(Q), \tau_{i+1}(Q') = \tau_{i+1}(Q)\},$$

$$\ell_i(Q) = \text{cl}(\mu(S_i(Q))), \quad \tau_i(Q) = (\ell_i(Q), \tau_{i+1}(Q)).$$

For a type $\tau = (\ell_0, \dots, \ell_k)$ let G_τ be the union of the heavy cells Q with $\tau_0(Q) = \tau$.

Claim: if P is an n_i -cell with $\mu(P \cap G_\tau) > 0$, then $P \cap G_\tau = S_i(Q)$ for every heavy $Q \subseteq P \cap G_\tau$; in particular $\text{cl}(\mu(P \cap G_\tau)) = \ell_i$. For $i = k$, P is a single heavy N -cell with $\tau_0(P) = \tau$, so $P \cap G_\tau = P = S_k(P)$. For $i < k$, fix such a Q and put $S = S_i(Q)$. For $Q' \in S$ we have $S_i(Q') = S$, so $\ell_i(Q') = \ell_i(Q)$ and $\tau_i(Q') = \tau_i(Q)$. For $j < i$, the cell $D_j(Q') = D_j(Q)$ contains P , so by downward induction on j , $\ell_j(Q') = \ell_j(Q)$. Hence $\tau_0(Q') = \tau$, that is, $S \subseteq G_\tau$. Conversely every heavy $Q' \subseteq P \cap G_\tau$ has $\tau_{i+1}(Q') = \tau_{i+1}(Q)$, so $Q' \subseteq S$.

Counting types. By the claim, $P \cap G_\tau$ is the union of the sets $P' \cap G_\tau$ over the at most 4^Δ cells P' of depth n_{i+1} inside P , each of class ℓ_{i+1} . So $2^{-w(\ell_{i+1}+1)} < \mu(P \cap G_\tau) \leq 4^\Delta 2^{-w\ell_{i+1}}$, which gives

$$\ell_{i+1} - \frac{2\Delta}{w} - 1 \leq \ell_i \leq \ell_{i+1}. \quad (5.1)$$

As $2\Delta/w \leq 8$, given ℓ_{i+1} there are at most 10 choices of ℓ_i . Hence the number of types is at most $(48/\varepsilon + 1) 10^{8/\varepsilon+1}$. By (P2) this is at most $R^{\kappa/4}$, since $\log_2((49/\varepsilon)10^{8/\varepsilon+1}) \leq \log_2(49/\varepsilon) + 3.33(8/\varepsilon + 1) \leq \kappa N/4$.

Discarding light types. Keep the types with $\mu(G_\tau) \geq R^{-\kappa/2}$. The discarded types have total mass at most $R^{\kappa/4} R^{-\kappa/2} = R^{-\kappa/4}$, so together with Step 0 at most $R^{-1} + R^{-\kappa/4} \leq 2R^{-\kappa/4}$ is discarded. (The threshold is tied to κ , not to ε : the mass of a part enters only through Lemma 5.10(iii), where a loss of order κ is affordable.)

Regularity. Let $\rho = \mu|_{G_\tau}/\mu(G_\tau)$ for a kept type, and let $n_i \leq n \leq n_{i+1}$. Every occupied n -cell of ρ contains an occupied n_{i+1} -cell, so its μ -mass in G_τ exceeds $2^{-w(\ell_{i+1}+1)}$. Every n -cell lies in an n_i -cell, so its μ -mass in G_τ is at most $2^{-w\ell_i}$. By (5.1), the ratio of the largest to the smallest occupied n -cell mass is at most $2^{w(\ell_{i+1}+1-\ell_i)} \leq 2^{2\Delta+2w} \leq 2^{3\varepsilon N/4} < R^\varepsilon$. $\square$

We apply Lemma 5.8 to μ_1 and to μ_2 . The normalized restrictions are the *regular pieces*; each lives on its *support set* G_i , a union of N -cells of positive μ_j -mass, with $\mu_j(G_i \setminus S_j) = 0$. Such cells meet S_j , so they lie within $\sqrt{2}2^{-N}$ of S_j ; as S_1 and S_2 are at distance at least 0.24, the support sets of pieces of μ_1 are disjoint from those of pieces of μ_2 .

5.5 The excess function

Definition 5.9 (excess function and heights). For a regular piece define $A(n)$ by $M(n) = 2^{-n} R^{-A(n)}$, so the heaviest n -cell has linear density $2^n M(n) = R^{-A(n)}$. For depths $i \leq p \leq j$ the *height* is $k_p[i, j] = A(p) - \min_{i \leq n \leq j} A(n) \geq 0$.

An occupied g -cell has mass at most $M(g)$, and an occupied p -cell at least $R^{-\varepsilon} M(p)$. Hence, for $g \leq p$,

$$\text{a } g\text{-cell contains at most } R^\varepsilon 2^{p-g} R^{A(p)-A(g)} \text{ occupied } p\text{-cells.} \quad (5.2)$$

Lemma 5.10 (properties of A). *For every regular piece of μ_1 or μ_2 , and $N \geq N_0$:*

- (i) $A(0) = 0$;
- (ii) $|A(n) - A(n')| \leq |n - n'|/N$;
- (iii) $A(n) \geq g(n) - \kappa$; in particular $A \geq -\kappa$ everywhere;
- (iv) $A(n) \leq n/N$.

Proof. (i) is $M(0) = 1$. (ii) holds because $M(n)/M(n+1) = 2R^{A(n+1)-A(n)} \in [1, 4]$, and (iv) follows from (i) and (ii). For (iii) with $n \geq 1$: an n -cell lies in a disc of radius $2^{-n} < 1$, so its μ_j -mass is at most $C_0 h_\theta(2^{-n}) = C_0 2^{-n} R^{-g(n)}$ by (2.1). The piece has μ_j -mass at least $R^{-\kappa/2}$, so after normalizing the cell has mass at most $C_0 R^{\kappa/2} 2^{-n} R^{-g(n)}$. Thus $R^{-A(n)} \leq C_0 R^{\kappa/2 - g(n)}$, and $\log_2 C_0 \leq \kappa N/2$ gives $A(n) \geq g(n) - \kappa$. $\square$

6 The filter

Throughout this section ρ is a regular piece with excess function A , and P is an occupied p -cell of ρ . The definitions are those of [7, §4]; only the constants are made explicit.

Remark 6.1 (use of (P1)). By (P1) and monotonicity, every number of the form $2^a(T+1)^b(N+16)^c$ with $a \leq 80$ and $b, c \leq 12$ is at most $R^{\varepsilon/8}$. We use this without comment, saying “by (P1)”.

6.1 Two tests

Definition 6.2 (tube test). Let $g < p$ and $1 \leq \lambda \leq 2$. The *tube of width λ* of P in direction w is the set of points within distance $\lambda 2^{-p} R^{2\varepsilon}$ of the line $c_P + \mathbb{R}w$ and within distance 2^{2-g} of c_P . Let $N_P^\lambda(w)$ be the number of occupied p -cells whose centers lie in it, and $\bar{N}_P = \frac{1}{2\pi} \int_{S^1} N_P^2(w) dw$. The direction w *passes the tube test $[g, p]$ at P at width λ* if $N_P^\lambda(w) \leq R^{2\varepsilon} \bar{N}_P$.

Definition 6.3 (projection test). Let $p < u$ and $1 \leq \lambda \leq 2$. Let $\rho_P = \rho|_P / \rho(P)$ and

$$\Pi_P^\lambda(w) = (\rho_P \times \rho_P) \{ (z, z') : |(z - z') \cdot w| \leq \lambda 2^{-u} R^{2\varepsilon} \}, \quad \bar{\Pi}_P = \frac{1}{2\pi} \int_{S^1} \Pi_P^2(w) dw.$$

The direction w *passes the projection test $[p, u]$ at P at width λ* if $\Pi_P^\lambda(w) \leq R^{2\varepsilon} \bar{\Pi}_P$.

For both tests p is the *anchor* and $p-g$, resp. $u-p$, is the *length*. Both tests are invariant under $w \mapsto -w$. The counts $N_P^\lambda(w)$ and $\Pi_P^\lambda(w)$ increase with λ , and they are upper semicontinuous in w (the tubes and the sets in the definition of Π are closed and depend continuously on w), hence Borel.

Lemma 6.4 (tests rarely fail). *For every test and every occupied anchor cell, the directions failing the test at width 2 form a set of arc length at most $2\pi R^{-2\varepsilon}$. A direction that passes at some width passes at every smaller width.*

Proof. Markov’s inequality, since $\bar{N}_P$ and $\bar{\Pi}_P$ are the averages of the width-2 counts; and monotonicity in λ . $\square$

Lemma 6.5 (average counts). *For the tube test $[g, p]$ and the projection test $[p, u]$ at P ,*

$$\bar{N}_P \leq R^{k_p[g,p]+4\epsilon}, \quad \bar{\Pi}_P \leq 2^{-(u-p)} R^{k_p[p,u]+4\epsilon}.$$

Consequently, if w passes the tube test $[g, p]$ at P at some width λ , then $N_P^\lambda(w) \leq R^{k_p[g,p]+6\epsilon}$; if w passes the projection test $[p, u]$ at P at width λ , then $\Pi_P^\lambda(w) \leq 2^{-(u-p)} R^{k_p[p,u]+6\epsilon}$.

Proof. Tube test. Let $Q \neq P$ be an occupied p -cell with $2^{-d-1} < |c_Q - c_P| \leq 2^{-d}$. Distinct centers are at least 2^{-p} apart, so $d \leq p$; the piece lies in $[0, 1)^2$, so $d \geq -1$; and if c_Q lies in some tube then $d \geq g - 2$. The directions w for which c_Q is within $2 \cdot 2^{-p} R^{2\epsilon}$ of $c_P + \mathbb{R}w$ satisfy $|\sin \angle(w, c_Q - c_P)| \leq 4 \cdot 2^{d-p} R^{2\epsilon}$, a set of arc length at most $8\pi 2^{d-p} R^{2\epsilon}$. Put $d' = \max(d, 0)$. The centers within 2^{-d} of c_P lie in the 3×3 block of d' -cells around the d' -cell of c_P , so by (5.2) there are at most $9R^\epsilon 2^{p-d'} R^{A(p)-A(d')}$ such Q . If $d' \geq g$ then $A(p) - A(d') \leq k_p[g, p]$; if $g - 2 \leq d' < g$ then $A(d') \geq A(g) - 2/N$ by Lemma 5.10(ii). Adding 2π for P itself and summing over at most $N + 4$ values of d ,

$$\int_{S^1} N_P^2(w) dw \leq 2\pi + (N + 4) \cdot 72\pi R^{3\epsilon} R^{k_p[g,p]+2/N},$$

so $\bar{N}_P \leq 150(N + 4)R^{3\epsilon} R^{k_p[g,p]} \leq R^{k_p[g,p]+4\epsilon}$ by (P1).

Projection test. Let $\eta = 2 \cdot 2^{-u} R^{2\epsilon}$. For $z \neq z'$, the directions with $|(z - z') \cdot w| \leq \eta$ form a set of arc length at most $2\pi \min(1, \eta/|z - z'|)$, so $\int \Pi_P^2 \leq 2\pi \iint \min(1, \eta/|z - z'|) d\rho_P d\rho_P$. A disc of radius 2^{-d} meets at most nine d -cells, and $\rho(P) \geq R^{-\epsilon} M(p)$, so for $d \geq p$

$$\rho_P(B(z, 2^{-d})) \leq 9R^\epsilon M(d)/M(p) = 9R^\epsilon 2^{-(d-p)} R^{A(p)-A(d)}.$$

Split the pairs into those with $|z - z'| \leq 2^{-u}$ and those with $2^{-d-1} < |z - z'| \leq 2^{-d}$ for $p - 1 \leq d < u$ (note $|z - z'| < 2^{1-p}$). For $d \geq p$ the part d contributes at most $4 \cdot 2^{d-u} R^{2\epsilon} \cdot 9R^\epsilon 2^{p-d} R^{A(p)-A(d)} = 36R^{3\epsilon} 2^{p-u} R^{A(p)-A(d)}$; the part $d = p - 1$ contributes at most $2 \cdot 2^{p-u} R^{2\epsilon}$; and the pairs with $|z - z'| \leq 2^{-u}$ contribute at most $9R^\epsilon 2^{p-u} R^{A(p)-A(u)}$. Since $A(p) - A(d) \leq k_p[p, u] + 1/N$ for $p - 1 \leq d \leq u$, summing gives $\bar{\Pi}_P \leq 75(N + 2)R^{3\epsilon} 2^{-(u-p)} R^{k_p[p,u]} \leq 2^{-(u-p)} R^{k_p[p,u]+4\epsilon}$.

The consequences follow from the definitions of passing. $\square$

6.2 Masks and levels

For a test of length ℓ put

$$\varsigma_\ell = \min(1, 2^{-\ell} R^\epsilon),$$

the angular scale of its mask.

Definition 6.6 (mask). Let a test have length ℓ and let $1 \leq \lambda' < \lambda \leq 2$. Let $Z \subseteq S^1$ be the set of directions that pass the test at P at width λ , and let Z_ς be its closed ς_ℓ -neighbourhood (in arc length). The *mask* of the test at P with widths (λ, λ') is $m_P = \mathbf{1}_{Z_\varsigma} * \omega_{\varsigma_\ell}$, the convolution being in the angle. For $x \in P$ we write $m(x, w) = m_P(w)$.

Lemma 6.7 (masks). m_P is smooth and even, $0 \leq m_P \leq 1$, $m_P = 1$ on Z , $m_P = 0$ outside $Z_{2\varsigma_\ell}$, and $|\partial_w^k m_P| \leq (\pi^2/6)^k (k!)^2 \varsigma_\ell^{-k}$. If $\lambda - \lambda' \geq 8R^{-\epsilon}$, then every direction in $Z_{2\varsigma_\ell}$ passes the test at P at width λ' ; in particular all derivatives of m_P vanish at every direction that fails the test at width λ' .

Proof. The first statements are Corollary 3.3; evenness ($m_P(-w) = m_P(w)$) holds because $Z = -Z$ and convolution in the angle commutes with the rotation by π . Let $w \in Z_{2\varsigma_\ell}$, that is, w is at arc distance at most $2\varsigma_\ell$ from Z . Since $\sin \varsigma_\ell < \varsigma_\ell$, there is $w_0 \in Z$ at arc distance less than $2\arcsin \varsigma_\ell$ from w , so that the chord satisfies $|w - w_0| < 2\varsigma_\ell$.

Tube test $[g, p]$, $\ell = p - g$. Let $z = c_P + tw + sw^\perp$ lie in the tube of width λ' in direction w , so $|t| \leq |z - c_P| \leq 2^{2-g}$ and $|s| \leq \lambda' 2^{-p} R^{2\varepsilon}$. Then

$$\text{dist}(z, c_P + \mathbb{R}w_0) \leq |t||w - w_0| + |s| \leq 2^{2-g} \cdot 2 \cdot 2^{g-p} R^\varepsilon + \lambda' 2^{-p} R^{2\varepsilon} \leq \lambda 2^{-p} R^{2\varepsilon},$$

using $\varsigma_\ell \leq 2^{-\ell} R^\varepsilon$ and $\lambda - \lambda' \geq 8R^{-\varepsilon}$. So the tube of width λ' in direction w lies in the tube of width λ in direction w_0 , and $N_P^{\lambda'}(w) \leq N_P^\lambda(w_0) \leq R^{2\varepsilon} \bar{N}_P$.

Projection test $[p, u]$, $\ell = u - p$. For $z, z' \in P$, $|(z - z') \cdot (w - w_0)| \leq \sqrt{2} 2^{-p} \cdot 2 \cdot 2^{p-u} R^\varepsilon < (\lambda - \lambda') 2^{-u} R^{2\varepsilon}$, so $\Pi_P^{\lambda'}(w) \leq \Pi_P^\lambda(w_0)$. $\square$

Definition 6.8 (levels). Put $\lambda_i = 2 - i/I_*$ for $0 \leq i \leq I_*$. A *level- i mask* is a mask with widths $(\lambda_i, \lambda_{i+1})$. By (2.2), $1/I_* \geq 16R^{-\varepsilon/8} \geq 8R^{-\varepsilon}$, so level- i masks exist for $i < I_*$.

Lemma 6.9 (three facts about levels).

- (L1) A level-0 mask equals 1 wherever the test passes at width 2.
- (L2) Wherever a level- i mask or one of its derivatives is nonzero, the test passes at width λ_{i+1} .
- (L3) If w_0 passes a test of length ℓ at P at width λ_i , then every w within arc distance $2^{-\ell} R^{\varepsilon/2}$ of w_0 passes it at P at width λ_{i+1} .

Proof. (L1) and (L2) are Lemma 6.7. For (L3), the computation in the proof of Lemma 6.7, with $2\varsigma_\ell$ replaced by $2^{-\ell} R^{\varepsilon/2}$, shows that the tube moves by at most $4 \cdot 2^{-p} R^{\varepsilon/2}$ and that $(z - z') \cdot w$ changes by at most $\sqrt{2} 2^{-u} R^{\varepsilon/2}$. Both are less than $2^{-p} R^{2\varepsilon}/I_*$, resp. $2^{-u} R^{2\varepsilon}/I_*$, because $4I_* \leq R^{3\varepsilon/2}$. $\square$

6.3 Symbols

Definition 6.10 (symbols). Fix a level $i < I_*$ and a finite list $\mathcal{L}$ of tests of ρ , of lengths at most L . For $t \in \mathcal{L}$ and x in the support set of ρ , let $m_t(x, w)$ be the level- i mask of t at the anchor cell of t containing x (it is occupied), and let

$$m_t^{[k]} = \frac{\varsigma_{\ell_t}^k \partial_w^k m_t}{(\pi^2/6)^k (k!)^2}, \quad \text{so that } |m_t^{[k]}| \leq 1, \quad m_t^{[0]} = m_t.$$

A *symbol of order at most k_0 built from $\mathcal{L}$* is a real function $b(x, w) = f(x) \prod_{t \in \mathcal{L}} m_t^{[k_t]}(x, w)$ with f Borel, $|f| \leq 1$, and integers $k_t \geq 0$ with $\sum_t k_t \leq k_0$. The symbol with $f \equiv 1$ and all $k_t = 0$ is the *product of the masks*. Put

$$M_{\mathcal{L}} = 2^7 T^2 |\mathcal{L}| \max(1, 2^L R^{-\varepsilon}).$$

Lemma 6.11 (properties of symbols). Let b be a symbol built from $\mathcal{L}$. Properties (S1), (S2) and (S4) hold for symbols of every order; (S3) is stated for symbols of order at most $2T$.

- (S1) $|b| \leq 1$.
- (S2) If $b(x, w) \neq 0$, then w passes every test of $\mathcal{L}$ at x at width λ_{i+1} .

(S3) For $1 \leq j \leq 6T$, $\partial_w^j b = \sum_i c_i b_i$, a finite sum of symbols b_i of order at most $2T + j$ built from $\mathcal{L}$ (with the same f), and $\sum_i |c_i| \leq M_{\mathcal{L}}^j$. In particular $b(x, \cdot)$ is $(1, M_{\mathcal{L}})$ -regular up to order $6T$.

(S4) If a test $t \in \mathcal{L}$ is anchored at a depth at most a , then each factor $m_t^{[k]}(x, w)$ is, on every a -cell, a function of w alone, bounded by 1.

Proof. (S1) holds because $|m_t^{[k]}| \leq 1$ for every k (Lemma 6.7), and (S4) is clear. (S2) follows from (L2): if $m_t^{[k]}(x, w) \neq 0$, then $\partial_w^k m_t(x, w) \neq 0$, so w lies in the support of the mask. For (S3), $\partial_w m_t^{[k]} = \varsigma_{\ell_t}^{-1}(\pi^2/6)(k+1)^2 m_t^{[k+1]}$, so differentiating a symbol of order k gives a sum of $|\mathcal{L}|$ symbols of order $k+1$ with coefficients at most $(\pi^2/6)(k+1)^2 / \min_t \varsigma_{\ell_t}$. Iterating j times from order at most $2T$, the coefficient sum is at most $(|\mathcal{L}|(\pi^2/6)(8T)^2 / \min_t \varsigma_{\ell_t})^j \leq M_{\mathcal{L}}^j$, since $1/\varsigma_{\ell} = \max(1, 2^\ell R^{-\varepsilon})$. $\square$

All lists below have at most $N^2 + 1$ tests, so by (2.2)

$$M_{\mathcal{L}} \leq 2^7 T^2 (N^2 + 1) \max(1, 2^L R^{-\varepsilon}) \leq R^{\varepsilon/8} \max(1, 2^L R^{-\varepsilon}).$$

6.4 The filtered measure

Let μ_1, μ_2 be as in Proposition 5.6 and let α be their distance measure. Suppose each regular piece ρ of μ_1 or μ_2 is given a list $\mathcal{L}(\rho)$ of at most $N^2 + 1$ tests of ρ ; the lists are fixed in §9. For x in the support set of a piece ρ , let $b(x, w)$ be the product of the level-0 masks of the tests of $\mathcal{L}(\rho)$ at x ; for x outside all support sets put $b(x, w) = 0$.

Definition 6.12 (filtered distance measure). τ_N is the image under $(x, y) \mapsto |x - y|$ of $|x - y|^{-1/2} b(x, e(x, y)) b(y, e(x, y)) d\mu_1(x) d\mu_2(y)$.

Lemma 6.13 (the filter removes summable mass). $0 \leq \tau_N \leq \alpha$, and

$$\alpha(\mathbb{R}) - \tau_N(\mathbb{R}) \leq a_N := 2.1 \left[4R^{-\kappa/4} + 2(3(N+1)R^{-\varepsilon} + 2^8 \mathcal{K}_0(\varepsilon N)^{-8}) \right].$$

Moreover $\sum_{N \geq N_0} a_N < \infty$.

Proof. Masks take values in $[0, 1]$, so $0 \leq \tau_N \leq \alpha$, and the weight is at most $0.24^{-1/2} < 2.1$. A pair (x, y) loses weight only if x or y lies outside all support sets, which concerns mass at most $2R^{-\kappa/4}$ for each of μ_1, μ_2 (Lemma 5.8), or if some test of the list of the piece of x fails at x at width 2 in direction $e(x, y)$ (by (L1)), or the same happens at y .

Fix $x \in S_1$, and let Z_x be the union, over the at most $N^2 + 1$ tests in the list of the piece of x , of the sets of directions failing the test at x at width 2. Then $Z_x = -Z_x$ and $|Z_x| \leq 2\pi(N^2 + 1)R^{-2\varepsilon}$ by Lemma 6.4. Since $e(x, y) = -\pi_x(y)$, the points y with $e(x, y) \in Z_x$ have μ_2 -mass $\int_{Z_x} F_x$. For $H > 1$,

$$\int_{Z_x} F_x \leq H|Z_x| + \int_{\{F_x > H\}} F_x \leq H|Z_x| + [\log H]^{-8} \int \Phi_8(F_x) \leq H|Z_x| + \mathcal{K}_0[\log H]^{-8}.$$

With $H = |Z_x|^{-1/2}$ we have $H|Z_x| \leq (2\pi(N^2 + 1))^{1/2} R^{-\varepsilon} \leq 3(N+1)R^{-\varepsilon}$, and $\log H \geq \varepsilon N \log 2 - \frac{1}{2} \log(2\pi(N^2 + 1)) \geq \varepsilon N/2$ by (P1). So the mass is at most $3(N+1)R^{-\varepsilon} + 2^8 \mathcal{K}_0(\varepsilon N)^{-8}$. Integrating in x , and arguing in the same way at y with the radial projections of μ_1 , gives the bound.

For summability: $R^{-\kappa/4} = 2^{-\kappa N/4}$ and $NR^{-\varepsilon} = N2^{-\varepsilon N}$ are summable since $\kappa N \geq \varepsilon N \asymp N^{2\theta-1}$; and $(\varepsilon N)^{-8} \asymp N^{-8(2\theta-1)}$, which is summable because $8(2\theta - 1) > 1$. $\square$

7 Two energies and four estimates

Setting. ρ is a regular piece with excess function A ; cells are cells of ρ , occupied unless said otherwise. Two a -cells X, Y are *separated* if $K2^{-a} \leq |c_X - c_Y| \leq 10K2^{-a}$, where $K = 1000$. Then $K2^{-a}/2 \leq |x - y| \leq 11K2^{-a}$ for $x \in X, y \in Y$, and all directions $e(x, y)$ lie in an arc of length less than $1/100$. Since each piece lies in a disc of radius $1/400$, separated cells exist only when $a \geq 10$. At depth $a = 0$ only, we also allow the pair (S_1, S_2) of Proposition 5.6, the first set carrying a regular piece ρ_1 of μ_1 and the second a regular piece ρ_2 of μ_2 , each with its own list; we call this the *root case*. All symbols are at a fixed level $i < I_* - 1$.

7.1 The two energies

Definition 7.1 (energy per unit mass). For a cell X (or S_j) carrying a symbol b , put

$$U_X(r, w) = \int_X e^{-irw \cdot x} b(x, w) d\rho(x), \quad \sigma_X(r) = \frac{r}{\rho(X)} \int_{S^1} |U_X(r, w)|^2 dw.$$

For two cells with symbols b_X, b_Y and a depth v , the *Fourier energy* is

$$F_{X,Y}(v) = \text{avg}_{r \approx 2^v} \sigma_X(r) \sigma_Y(r).$$

- *Trivial bound.* Since $|b| \leq 1$, $\sigma_X(r) \leq 2\pi r \rho(X)$; with $\text{avg } r^2 \leq 64 \cdot 4^v$ and $\rho(X) \leq M(a) = 2^{-a} R^{-A(a)}$, for a -cells

$$F_{X,Y}(v) \leq 2600 \cdot 4^v \rho(X) \rho(Y) \leq 2600 \cdot 4^{v-a} R^{-2A(a)}. \quad (7.1)$$

- *Reached masks can be dropped.* By (S4), a factor of b whose test is anchored at a depth at most a is a function of w alone on each a -cell, bounded by 1; it comes out of U_X and can only decrease $|U_X|$. So in an upper bound for F at depth a such factors may be removed, and the result is a symbol built from the remaining tests. Such tests are *reached* at depth a .

Definition 7.2 (distance energy). Let X, Y be separated a -cells, $\mathcal{L}$ a list of tests, and i a level. Let Γ be the set of $(x, y) \in X \times Y$ such that $e(x, y)$ passes every test of $\mathcal{L}$ at x and at y at width λ_i . Let $\nu_{X,Y}$ be the image of $|x - y|^{-1/2} \mathbf{1}_\Gamma d\rho(x) d\rho(y)$ under $(x, y) \mapsto |x - y|$, and

$$D_{X,Y}(t) = \frac{2^t}{\rho(X) \rho(Y)} (\nu_{X,Y} \times \nu_{X,Y}) \{(s, s') : |s - s'| \leq 2^{-t}\}.$$

- *Trivial bound.* The weight is at most $(2^{a+1}/K)^{1/2}$, so

$$D_{X,Y}(t) \leq 2^{t+a} \rho(X) \rho(Y) \leq 2^{t-a} R^{-2A(a)}. \quad (7.2)$$

- *Positivity.* Removing tests from the list, or computing at a higher level (a smaller width), enlarges Γ and can only increase D .

Lemma 7.3 (binning). Let η, ζ be finite positive measures on $\mathbb{R}$ and $h > 0$. With absolute constants:

$$(i) \quad (\eta \times \eta) \{|s - s' - u| \leq h\} \leq 4 (\eta \times \eta) \{|s - s'| \leq h\} \text{ for every } u \in \mathbb{R};$$

- (ii) $(\eta \times \eta)\{|s - s'| \leq Lh\} \leq (2L + 1)(\eta \times \eta)\{|s - s'| \leq h\}$ for every integer $L \geq 1$;
- (iii) $\iint (1 + |s - s'|/h)^{-2} d\eta(s) d\zeta(s') \leq 10((\eta \times \eta)\{|s - s'| \leq h\} \cdot (\zeta \times \zeta)\{|s - s'| \leq h\})^{1/2}$;
- (iv) $(\eta \times \eta)\{|s - s'| \leq h\} \leq 2h \int_{|r| \leq 1/h} |\hat{\eta}(r)|^2 dr$, and $\int_{|r| \leq 1/h} |\hat{\eta}(r)|^2 dr \leq 160 h^{-1}(\eta \times \eta)\{|s - s'| \leq h\}$.

Proof. Let $b_j = \eta([jh, (j+1)h))$ and $c_j = \zeta([jh, (j+1)h))$; then $\sum_j b_j^2 \leq (\eta \times \eta)\{|s - s'| \leq h\}$. If s is in bin j and s' in bin $j - m$, then $(m-1)h < s - s' < (m+1)h$. So the left side of (i) is at most a sum of $\sum_j b_j b_{j-m}$ over the at most four integers $m \in (u/h - 2, u/h + 2)$, and by Cauchy–Schwarz each such sum is at most $\sum_j b_j^2$. (ii) is the same with the $2L + 1$ shifts $|m| \leq L$. In (iii) the pairs in bins $j, j - m$ have weight at most m^{-2} when $|m| \geq 1$, and summing over m with Cauchy–Schwarz gives the constant $1 + 2 \sum_{m \geq 1} m^{-2} < 10$. For the first part of (iv) use the Fejér kernel $\mathcal{F}(s) = (\sin(s/2h)/(s/2h))^2$, which is at least 0.9 on $[-h, h]$ and whose Fourier transform $2\pi h(1 - h|r|)_+$ is nonnegative and supported in $|r| \leq 1/h$; by Parseval, $0.9(\eta \times \eta)\{|s - s'| \leq h\} \leq \iint \mathcal{F}(s - s') d\eta d\eta = \frac{1}{2\pi} \int 2\pi h(1 - h|r|)_+ |\hat{\eta}(r)|^2 dr$. For the second part, $\mathbf{1}_{|r| \leq 1/h} \leq (2 - h|r|)_+$, and

$$\int (2 - h|r|)_+ |\hat{\eta}(r)|^2 dr = \iint \frac{4 \sin^2((s - s')/h)}{h((s - s')/h)^2} d\eta(s) d\eta(s') \leq \frac{16}{h} \iint \left(1 + \frac{|s - s'|}{h}\right)^{-2} d\eta d\eta,$$

because $4 \min(1, u^{-2}) \leq 16(1 + |u|)^{-2}$. Now (iii) with $\zeta = \eta$ gives the constant 160. $\square$

Lemma 7.4 (low and high frequencies). *Let X, Y be separated a -cells, $\mathcal{L}$ a list, i a level, and b the product of the level- i masks of $\mathcal{L}$. Let $\nu_{X,Y}^b$ be the image under $(x, y) \mapsto |x - y|$ of the measure*

$$|x - y|^{-1/2} b(x, e(x, y)) b(y, e(x, y)) d\rho(x) d\rho(y) \quad \text{on } X \times Y.$$

If $t - \varepsilon N > a$, then

$$D_{X,Y}(t) \leq C_1 D_{X,Y}^+(t - \varepsilon N) + \frac{C_1}{\rho(X)\rho(Y)} \sum_{t - \varepsilon N < v \leq t} \int_{2^{v-1} \leq |r| \leq 2^v} |\hat{\nu}_{X,Y}^b(r)|^2 dr,$$

where D^+ is computed with $\mathcal{L}$ at level $i + 1$ and $C_1 = 320$.

Proof. Each level- i mask equals 1 where its test passes at width λ_i (Lemma 6.7). So the product $b(x, e(x, y)) b(y, e(x, y))$ equals 1 for $(x, y) \in \Gamma$; and $b \geq 0$. Hence $\nu_{X,Y} \leq \nu_{X,Y}^b$, and by Lemma 7.3(iv),

$$D_{X,Y}(t) \leq \frac{2}{\rho(X)\rho(Y)} \int_{|r| \leq 2^t} |\hat{\nu}_{X,Y}^b(r)|^2 dr.$$

Split at $|r| = 2^{t - \varepsilon N}$. By the second part of Lemma 7.3(iv) the low part is at most

$$\frac{2 \cdot 160 \cdot 2^{t - \varepsilon N}}{\rho(X)\rho(Y)} (\nu^b \times \nu^b)\{|s - s'| \leq 2^{-(t - \varepsilon N)}\};$$

and ν^b is at most the measure of Definition 7.2 at level $i + 1$, because $0 \leq b \leq 1$ and b vanishes unless every test passes at width λ_{i+1} (L2). The high part is the sum over v . $\square$

7.2 Estimate I: Mattila's identity with masks

Estimate 7.5 (Mattila's identity with masks). *Let X, Y be separated a -cells, or the root case $a = 0$, $(X, Y) = (S_1, S_2)$. Let b_X, b_Y be the products of the level- i masks of lists $\mathcal{L}_X, \mathcal{L}_Y$ (equal to one list $\mathcal{L}$ for cells of one piece). Let ν^b be the image under $(x, y) \mapsto |x - y|$ of*

$$|x - y|^{-1/2} b_X(x, e(x, y)) b_Y(y, e(x, y)) d\rho_X(x) d\rho_Y(y) \quad \text{on } X \times Y.$$

Suppose

$$v \geq a + 10\varepsilon N \quad \text{and every test of } \mathcal{L}_X \cup \mathcal{L}_Y \text{ has length } \ell \text{ with } 2\ell \leq v - a + \varepsilon N.$$

Then

$$\frac{1}{\rho_X(X)\rho_Y(Y)} \int_{2^{v-1} \leq |r| \leq 2^{v+1}} |\nu^b(r)|^2 dr \leq 2 \sup F_{X,Y}(v) + R^{-30},$$

where the supremum is over pairs of real symbols of order at most $2T$ built from $\mathcal{L}_X$ (for X) and $\mathcal{L}_Y$ (for Y) at level i .

Proof. Step 0: geometry and scales. In both cases $X \subseteq B(x_*, \varrho)$, $Y \subseteq B(y_*, \varrho)$ with $|x_* - y_*| \geq 50\varrho$: for cells take the centers and $\varrho = 2^{-a}/\sqrt{2}$; in the root case take x_0, y_0 and $\varrho = 1/400$. In both cases $0.24 \cdot 2^{-a} \leq |x - y|$, and $\sqrt{2}/|x_* - y_*| \leq 6 \cdot 2^a$. All directions $e(x, y)$ lie in an arc Γ of length at most $1/20$; let $\psi = \psi_\Gamma$, which equals 1 on $\Gamma_{0.1}$ and vanishes on the $(\pi/3)$ -neighbourhood of $-\Gamma$.

Let $M = \max(M_{\mathcal{L}_X}, M_{\mathcal{L}_Y})$. Since $2\ell \leq v - a + \varepsilon N$ and $v - a \geq 10\varepsilon N$, $\max(1, 2^\ell R^{-\varepsilon}) \leq 2^{(v-a)/2} R^{-\varepsilon/2}$, hence

$$\begin{aligned} M &\leq 2^7 T^2 (N^2 + 1) 2^{(v-a)/2} R^{-\varepsilon/2}, \\ M_* &:= 2M + 1200 T^2 \leq 2^{11} T^2 (N^2 + 1) 2^{(v-a)/2} R^{-\varepsilon/2}. \end{aligned} \tag{7.3}$$

Fix $x \in X$, $y \in Y$, and write $z = x - y$, ϕ_z for the angle of $e(x, y) = z/|z|$, and $B(w) = b_X(x, w) b_Y(y, w)$. By (S3) and the Leibniz rule, B is $(1, 2M)$ -regular up to order $6T$. Since $|\psi^{(k)}| \leq 33^k (k!)^2 \leq (1200 T^2)^k$ for $k \leq 6T$, for every polynomial $Q(\partial)$ of degree at most $2T$ the function $\psi \cdot QB$ is $(\|Q\|_{2M}, M_*)$ -regular up to order $4T$. For $r \in [2^{v-1}, 2^{v+1}]$ put $\Lambda = r|z| \geq 0.12 \cdot 2^{v-a} \geq 2^{v-a-4}$ and

$$\varpi = \frac{10^8 (T+1)^4 M_*^2}{\Lambda} \leq 2^{80} (T+1)^8 (N+16)^4 R^{-\varepsilon} \leq R^{-7\varepsilon/8}$$

by (7.3) and (P1).

Step 1: inversion. Apply Lemma 3.7 to $G_j = \psi \cdot Q_j B$ ($j < T$), with $\phi_0 = \phi_z$. Since G_j vanishes near $\phi_z + \pi$, the second sum there vanishes; since $\psi = 1$ near ϕ_z , $(P_i G_j)(\phi_z) = (P_i Q_j B)(\phi_z)$. Multiply by $r^{1/2} \Lambda^{-j}$, sum over $j < T$, and use $\sum_{i+j=m} P_i Q_j = 0$ for $1 \leq m < T$ (Lemma 3.8). With $c_1 = (2\pi)^{-1/2} e^{-i\pi/4}$ this gives

$$|z|^{-1/2} e^{-i\Lambda} B(\phi_z) = c_1 \sum_{j < T} \Lambda^{-j} r^{1/2} \int_{S^1} e^{-irw \cdot z} \psi(w) (Q_j B)(w) dw + \text{Err}, \tag{7.4}$$

where Err collects two kinds of terms. The terms $|z|^{-1/2} \Lambda^{-m} (\sum_{i+j=m, i, j < T} P_i Q_j B)(\phi_z)$ with $T \leq m \leq 2T - 2$ are at most $|z|^{-1/2} (m+1) (800 m M^2 / \Lambda)^m \leq |z|^{-1/2} (m+1) \varpi^m$ by Lemma 3.8.

The remainders $r^{1/2}\Lambda^{-j}E_j$ of Lemma 3.7 satisfy $\Lambda^{-j}\|Q_j\|_{2M} \leq 1$ and $|E_j| \leq \|Q_j\|_{2M} \varpi^T \cdot 10^8(T+1)^4 M_*^2$. Since $|z|^{-1/2} \leq 4R^{1/2}$, $r^{1/2} \leq R$, $M_*^2 \leq R^2$ and $\varpi^T \leq R^{-7\varepsilon T/8} \leq R^{-210}$, we get $|\text{Err}| \leq R^{-200}$.

Step 2: integrate and separate variables. Integrating (7.4) against $d\rho_X(x) d\rho_Y(y)$, the left side gives $\hat{\nu}^b(r)$. In the term j write $\Lambda^{-j} = r^{-j}|x-y|^{-j}$. For $j \geq 1$ expand $|x-y|^{-j} = \sum_k c_k^{(j)} f_k(x) g_k(y)$ by Lemma 3.9, with $\sum_k |c_k^{(j)}| \leq 2(6 \cdot 2^a)^j$; expand $Q_j B = \sum_l q_{jl} \sum_{s \leq l} \binom{l}{s} \partial^s b_X(x, \cdot) \partial^{l-s} b_Y(y, \cdot)$; and expand each derivative by (S3). The result is

$$\hat{\nu}^b(r) = c_1 \sum_{\alpha} \gamma_{\alpha}(r) Z_{\alpha}(r) + \iint \text{Err}, \quad Z_{\alpha}(r) = r^{1/2} \int_{S^1} \psi(w) U_X^{(\alpha)}(r, w) \overline{U_Y^{(\alpha)}(r, w)} dw,$$

where $U_X^{(\alpha)}$ and $U_Y^{(\alpha)}$ are computed with real symbols of order at most $2T$ built from $\mathcal{L}_X$, resp. $\mathcal{L}_Y$ (the spatial factors are the polynomials $f_k \mathbf{1}_X, g_k \mathbf{1}_Y$), and the term $j = 0$ is Z_0 with $\gamma_0 = 1$ and the symbols b_X, b_Y themselves. For $j \geq 1$ the coefficients satisfy, uniformly in $r \in [2^{v-1}, 2^{v+1}]$,

$$\sum_{\alpha \in \text{term } j} |\gamma_{\alpha}| \leq 2(12 \cdot 2^{a-v})^j \|Q_j\|_{2M} \leq 2(9600 T M^2 2^{a-v})^j \leq 2R^{-7\varepsilon j/8},$$

because $M^2 2^{a-v} \leq 2^{14} T^4 (N^2 + 1)^2 R^{-\varepsilon}$ by (7.3). Hence $\sum_{\alpha} \sup_r |\gamma_{\alpha}| \leq 1.1$.

Step 3: Cauchy-Schwarz. By Cauchy-Schwarz in w and $0 \leq \psi \leq 1$,

$$|Z_{\alpha}(r)|^2 \leq \rho_X(X) \rho_Y(Y) \sigma_X^{(\alpha)}(r) \sigma_Y^{(\alpha)}(r) / r.$$

Also $|\iint \text{Err}| \leq R^{-200} \rho_X(X) \rho_Y(Y)$. Hence, by Cauchy-Schwarz in α ,

$$|\hat{\nu}^b(r)|^2 \leq 2|c_1|^2 (1.1) \sum_{\alpha} \sup |\gamma_{\alpha}| \rho_X(X) \rho_Y(Y) \frac{\sigma_X^{(\alpha)}(r) \sigma_Y^{(\alpha)}(r)}{r} + 2R^{-400} (\rho_X(X) \rho_Y(Y))^2.$$

Integrate over $2^{v-1} \leq r \leq 2^{v+1}$ and use $\int \sigma \sigma r^{-1} dr \leq 2 \text{avg}_{r \approx 2^v} \sigma \sigma$. Since $|c_1|^2 = 1/(2\pi)$, the positive frequencies contribute at most $\frac{2 \cdot 1 \cdot 21 \cdot 2}{2\pi} \rho_X(X) \rho_Y(Y) \sup F + R^{-390} (\rho_X(X) \rho_Y(Y))^2$; the negative frequencies contribute the same, because $\hat{\nu}^b(-r) = \overline{\hat{\nu}^b(r)}$. Divide by $\rho_X(X) \rho_Y(Y) \leq 1$. $\square$

7.3 Estimate II: orthogonality

Estimate 7.6 (orthogonality). *Let X, Y be a -cells of ρ (any two, possibly equal), carrying symbols of order at most $2T$ built at level i from a list $\mathcal{L}$ whose tests have lengths at most L . Let $a < p \leq v$ with*

$$(H1) \quad v - p \geq 10\varepsilon N;$$

$$(H2) \quad p - a \leq v - p + 2\varepsilon N;$$

$$(H3) \quad L \leq v - p + 2\varepsilon N.$$

Suppose $\mathcal{L}$ contains a tube test $[g, p]$ with $a \leq g \leq a + \varepsilon N$. Then

$$F_{X,Y}(v) \leq R^{k_p[g,p]+12\varepsilon} \sum_{P,Q} \frac{\rho(P)\rho(Q)}{\rho(X)\rho(Y)} F_{P,Q}(v) + R^{-80},$$

the sum over occupied p -cells $P \subseteq X, Q \subseteq Y$, with the same symbols restricted to P and Q .

Proof. Step 1: expand. Since $\rho(X)\rho(Y)F_{X,Y}(v) = \text{avg}_{r \approx 2^v} r^2 \int |U_X(r, w_1)|^2 dw_1 \int |U_Y(r, w_2)|^2 dw_2$ and $U_X = \sum_{P \subseteq X} U_P$, $U_Y = \sum_{Q \subseteq Y} U_Q$, we get a finite sum over *rows* (P, Q) and *columns* (P', Q') of

$$\text{avg}_{r \approx 2^v} r^2 \iint U_P(r, w_1) \overline{U_{P'}(r, w_1)} U_Q(r, w_2) \overline{U_{Q'}(r, w_2)} dw_1 dw_2.$$

Step 2: arcs and links. Let $\ell' = \min(1, 2^{-(p-a)} R^\varepsilon)$, cut the circle into $\lceil 2\pi/\ell' \rceil$ equal arcs A_k (of length between $\ell'/2$ and ℓ') with centers w_k^0 , and let $\chi_k = \mathbf{1}_{A_k} * \omega_{|A_k|/4}$. Then $\sum_k \chi_k = 1$, χ_k vanishes at arc distance more than ℓ' from w_k^0 , and χ_k is $(1, 576T^2/\ell')$ -regular up to order $6T$ (Corollary 3.3). Insert $1 = \sum_{k_1} \chi_{k_1}(w_1) \sum_{k_2} \chi_{k_2}(w_2)$. Put $\tau = 2^{-p} R^{3\varepsilon/2}$ and call a row and a column *linked* for the arcs (k_1, k_2) if

- (i) $|w_{k_1}^{0\perp} \cdot (c_{P'} - c_P)| \leq \tau$ and $|w_{k_2}^{0\perp} \cdot (c_{Q'} - c_Q)| \leq \tau$;
- (ii) $|w_{k_1}^0 \cdot (c_P - c_{P'}) + w_{k_2}^0 \cdot (c_Q - c_{Q'})| \leq \tau$.

For points $x \in P$, $x' \in P'$, $y \in Q$, $y' \in Q'$ and w_j in the support of χ_{k_j} , replacing points by centers changes the quantities in (i), (ii) by at most $\sqrt{2} 2^{-p}$ per point pair, and replacing w_j by $w_{k_j}^0$ changes them by at most $\ell' \sqrt{2} 2^{-a} \leq \sqrt{2} 2^{-p} R^\varepsilon$ per pair. Since $R^{\varepsilon/2} \geq 12$ (by (P1)), if a condition fails at the centers by more than τ , the corresponding quantity at the points exceeds $\tau/2$.

We use the following bounds, all consequences of the hypotheses: by (H3) and (S3), the symbols are $(1, M)$ -regular in w with $M \leq 2^7 T^2 (N^2 + 1) 2^{v-p} R^\varepsilon$; by (H2), $576T^2/\ell' \leq 576T^2 2^{v-p} R^\varepsilon$; so the angular amplitude $\chi_k(w)b(x, w)b(x', w)$ is $(1, M_a)$ -regular up to order $4T$ with $M_a \leq 2^{10} T^2 (N^2 + 1) 2^{v-p} R^\varepsilon$.

If (ii) fails, the phase $r\Phi$, $\Phi = w_1 \cdot (x - x') + w_2 \cdot (y - y')$, has $|\Phi| \geq \tau/2$. The amplitude $h(r) = 2^{-v} \Psi(r/2^v) r^2$ is supported in $[2^{v-2}, 2^{v+2}]$ and satisfies $|h^{(j)}| \leq 2^{v+4} (512T^2 2^{-v})^j$ for $j \leq 6T$. The phase is linear in r , so the linear-phase case of Lemma 3.4 (on the interval $[2^{v-3}, 2^{v+3}]$, with $k = T$, $\lambda = \tau/2$, $A' = 2^{v+4}$ and $M' = 512T^2 2^{-v}$, which may be less than 1) gives

$$\left| \int e^{-ir\Phi} h(r) dr \right| \leq 2^{2v+6} \left(\frac{2^{10} T^2 2^{-v}}{\tau} \right)^T \leq 2^{2v+6} (2^{10} T^2 R^{-11.5\varepsilon})^T \leq R^{-300},$$

using $2^{-v}/\tau = 2^{p-v} R^{-3\varepsilon/2} \leq R^{-11.5\varepsilon}$ by (H1), and (P1).

If (i) fails, say for the first pair, integrate in the angle ϕ_1 of w_1 . The phase is $F = -rw_1 \cdot (x - x')$, so $F' = -rw_1^\perp \cdot (x - x')$ and $|F'| \geq r\tau/2$ on the support, and $\geq \lambda := r\tau/3$ on a neighbourhood of it; $|F^{(j)}| \leq r|x - x'| =: \Lambda_F$. Since $r \geq 2^{v-2}$, $\lambda \geq 2^{v-p-4} R^{3\varepsilon/2}$, and $\Lambda_F/\lambda = 3|x - x'|/\tau \leq 5 \cdot 2^{p-a} R^{-3\varepsilon/2} \leq 5 \cdot 2^{v-p} R^{\varepsilon/2}$ by (H2). Lemma 3.4 with $k = T$ gives

$$\begin{aligned} \left| \int \chi_{k_1}(w_1) e^{-irw_1 \cdot (x - x')} b(x, w_1) b(x', w_1) dw_1 \right| &\leq 2\ell' \left(\frac{2(T+1)^2 \cdot 2^{10} T^2 (N^2 + 1) 2^{v-p} R^\varepsilon}{2^{v-p-4} R^{3\varepsilon/2}} \right)^T \\ &\leq 2\ell' R^{-3\varepsilon T/8} \leq 2\ell' R^{-90}, \end{aligned}$$

by (P1). The second pair is the same.

In all cases, bounding the remaining integrals trivially, a term with unlinked row and column for the arcs (k_1, k_2) is at most $64 \cdot 4^v \cdot (2\ell')^2 R^{-90} \rho(P)\rho(P')\rho(Q)\rho(Q')$. There are at most $8/\ell'$ arcs, so summing over pairs of arcs and over all rows and columns, the unlinked terms contribute at most $2 \cdot 10^4 \cdot 4^v R^{-90} \rho(X)^2 \rho(Y)^2 \leq R^{-80} \rho(X)\rho(Y)$.

Step 3: few links. Fix a row (P, Q) and arcs (k_1, k_2) . A column contributes zero unless $b(x', w_1) \neq 0$ for some $x' \in P'$ and some w_1 in the support of χ_{k_1} , and similarly for Q' ; discard

the other columns. Group the remaining P' by their g -cell ancestor in X ; there are at most $4^{g-a} \leq R^{2\varepsilon}$ groups. In one group choose P_* with a witness (x_*, w_*) , $x_* \in P_*$, $b(x_*, w_*) \neq 0$. By (S2), w_* passes the tube test $[g, p]$ at P_* at width λ_{i+1} . Every P' of the group satisfies (i), so $c_{P'}$ and c_{P_*} lie within τ of the same line in direction $w_{k_1}^0$, and $|c_{P'} - c_{P_*}| \leq \sqrt{2} 2^{-g}$. Hence

$$\text{dist}(c_{P'}, c_{P_*} + \mathbb{R}w_*) \leq 2\tau + \sqrt{2} 2^{-g} \ell' \leq 2^{-p} (2R^{3\varepsilon/2} + \sqrt{2}R^\varepsilon) < 2^{-p} R^{2\varepsilon},$$

using $g \geq a$; and $|c_{P'} - c_{P_*}| < 2^{2-g}$. So $c_{P'}$ lies in the tube of width λ_{i+1} of P_* in direction w_* , and by Lemma 6.5 each group has at most $R^{k_p[g,p]+6\varepsilon}$ members. Once P' is fixed, (ii) and (i) put $c_{Q'}$ in a square of side 2τ , which contains at most $(3R^{3\varepsilon/2} + 1)^2 \leq 16R^{3\varepsilon}$ centers of p -cells. So every row is linked to at most $\mathbf{n} = R^{2\varepsilon} \cdot R^{k_p[g,p]+6\varepsilon} \cdot 16R^{3\varepsilon} \leq R^{k_p[g,p]+12\varepsilon}$ nonzero columns, and by symmetry every column to at most $\mathbf{n}$ nonzero rows.

Step 4: Schur's test. For fixed (k_1, k_2, r, w_1, w_2) put $F_{(P,Q)} = U_P(r, w_1)U_Q(r, w_2)$. Then $|\sum_{\text{linked}} F_\alpha \overline{F_\beta}| \leq \frac{1}{2} \sum_{\text{linked}} (|F_\alpha|^2 + |F_\beta|^2) \leq \mathbf{n} \sum_\alpha |F_\alpha|^2$. Integrate against $\chi_{k_1}(w_1)\chi_{k_2}(w_2) \geq 0$, sum over the arcs ($\sum \chi_k = 1$), and average r^2 :

$$\rho(X)\rho(Y)F_{X,Y}(v) \leq \mathbf{n} \sum_{P,Q} \rho(P)\rho(Q)F_{P,Q}(v) + R^{-80}\rho(X)\rho(Y). \quad \square$$

7.4 Estimate III: linearization

Estimate 7.7 (linearization). *Let X, Y be separated a -cells and $a < p < t$ with $2p > a + t$. Suppose the list $\mathcal{L}$ of $D_{X,Y}(t)$ (at level i) contains the projection test $[p, t]$, and let $\mathcal{L}' \subseteq \mathcal{L}$ consist of tests anchored at depths at most p and of lengths at most $(p - a)/2$. Then*

$$D_{X,Y}(t) \leq R^{k_p[p,t]+7\varepsilon} D'_{X,Y}(p),$$

where D' is computed with $\mathcal{L}'$ at level $i + 1$.

Proof. Call a pair (P, Q) of p -cells, $P \subseteq X$, $Q \subseteq Y$, *retained* if some $(x_0, y_0) \in P \times Q$ lies in Γ . Only retained pairs contribute to $\nu_{X,Y}$.

Linearization. Let $e_0 = e(c_P, c_Q)$. The Hessian of $z \mapsto |z|$ has norm $1/|z| \leq 2^{a+1}/K$ on the relevant segment, so for $x \in P$, $y \in Q$,

$$|x - y| = |c_P - c_Q| + e_0 \cdot ((x - c_P) - (y - c_Q)) + E, \quad |E| \leq \frac{1}{2} \cdot \frac{2^{a+1}}{K} \cdot (\sqrt{2} 2^{-p})^2 \leq \frac{2^{-t}}{K},$$

as $2p \geq a + t + 1$. So two distances from $P \times Q$ agree within 2^{-t} only if $|e_0 \cdot (x - x') - e_0 \cdot (y - y')| \leq 2 \cdot 2^{-t}$.

One cell pair. The direction $e(x_0, y_0)$ passes $[p, t]$ at Q at width λ_i , and $|e(x_0, y_0) - e_0| \leq 4\sqrt{2} 2^{a-p}/K < 2^{-(t-p)}$, because $t - p < p - a$. By (L3), e_0 passes $[p, t]$ at Q at width $\lambda_{i+1} \geq 1$, and by Lemma 6.5, $\Pi_Q^1(e_0) \leq \mathcal{B} := 2^{-(t-p)} R^{k_p[p,t]+6\varepsilon}$. Fix $x, x' \in P$. By Lemma 7.3(i) applied to the projection of $\rho|_Q$ on e_0 , and since $2 \cdot 2^{-t} \leq 2^{-t} R^{2\varepsilon}$, the pairs $y, y' \in Q$ with $|e_0 \cdot (y - y') - e_0 \cdot (x - x')| \leq 2 \cdot 2^{-t}$ have $\rho \times \rho$ -mass at most $4\rho(Q)^2 \Pi_Q^1(e_0) \leq 4\rho(Q)^2 \mathcal{B}$. Integrating over x, x' , the image η_{PQ} of $\rho|_P \times \rho|_Q$ under $(x, y) \mapsto |x - y|$ satisfies $(\eta_{PQ} \times \eta_{PQ})\{|s - s'| \leq 2^{-t}\} \leq 4\mathcal{B}\rho(P)^2\rho(Q)^2$.

All cell pairs. Group the retained pairs by the interval $[k2^{-p}, (k+1)2^{-p})$ containing $|c_P - c_Q|$. Distances in group k lie within $\sqrt{2} 2^{-p}$ of that interval, so distances from groups k, k' agree within 2^{-t} only if $|k - k'| \leq 4$. Cut the line into intervals of length 2^{-t} ; for a group G the vector of masses of $\eta_G = \sum_{(P,Q) \in G} \eta_{PQ}$ has ℓ^2 norm at most $\sum_{(P,Q) \in G} (4\mathcal{B})^{1/2} \rho(P)\rho(Q)$ by the triangle

inequality. As in the proof of Lemma 7.3(i) (pairs at distance at most 2^{-t} lie in the same or adjacent intervals), with Cauchy–Schwarz over the three adjacent intervals and over the at most nine neighbouring groups,

$$(\eta \times \eta)\{|s - s'| \leq 2^{-t}\} \leq C\mathcal{B} \sum_G W_G^2, \quad W_G = \sum_{(P,Q) \in G} \rho(P)\rho(Q),$$

where η is the image of $\rho \times \rho$ restricted to the retained pairs. Every pair (x, y) in a retained (P, Q) passes the tests of $\mathcal{L}'$ at width λ_{i+1} : the witness passes them at width λ_i ; their anchor cells contain P and Q , so the tests at x, y are those at x_0, y_0 ; and $e(x, y)$ is within arc distance $2^{a-p+1} \leq 2^{-\ell} R^{\varepsilon/2}$ of $e(x_0, y_0)$ for every $\ell \leq (p-a)/2$ (L3). All distances in a group lie in an interval of length $4 \cdot 2^{-p}$, so by Lemma 7.3(ii), $\sum_G W_G^2 \leq 9(\eta' \times \eta')\{|s - s'| \leq 2^{-p}\}$, where η' is the image of $\rho \times \rho$ restricted to the pairs passing $\mathcal{L}'$ at width λ_{i+1} . The weight $|x - y|^{-1/2}$ lies between $(11K2^{-a})^{-1/2}$ and $(2^{a+1}/K)^{1/2}$, so

$$D_{X,Y}(t) \leq \frac{2^t}{\rho(X)\rho(Y)} \cdot \frac{2^{a+1}}{K} \cdot C\mathcal{B} \cdot 9(\eta' \times \eta')\{|s - s'| \leq 2^{-p}\} \leq C' 2^{t-p} \mathcal{B} D'_{X,Y}(p),$$

with $C = 108$ (3 neighbouring bins, 9 neighbouring groups, and the factor 4) and $C' = 2 \cdot 108 \cdot 9 \cdot 11 = 21384 < 2^{15}$. Finally $C' 2^{t-p} \mathcal{B} = C' R^{k_p[p,t]+6\varepsilon} \leq R^{k_p[p,t]+7\varepsilon}$ by (P1). $\square$

7.5 Estimate IV: splitting in space

Estimate 7.8 (splitting in space). *Let X, Y be a -cells carrying symbols of order at most $2T$ built at level i from a list $\mathcal{L}$ whose tests have lengths at most L . Put $h = v - 2\max(L, \varepsilon N)$ and suppose $h > a$. Then*

$$F_{X,Y}(v) \leq 81^2 \sum_{P,Q} \frac{\rho(P)\rho(Q)}{\rho(X)\rho(Y)} F_{P,Q}(v) + C_{\text{IV}} \sum_{a < n < h+12} \max D_{P,P'}(v) + R^{-30}.$$

In the first sum $P \subseteq X$, $Q \subseteq Y$ are occupied h -cells, with the same symbols. In the second, for each n we may choose any sublist $\mathcal{L}_n \subseteq \mathcal{L}$ (possibly depending on n); the maximum is over separated pairs of occupied n -cells P, P' , both in X or both in Y , and $D_{P,P'}(v)$ is computed with $\mathcal{L}_n$ at level $i+1$. The constant C_{IV} is absolute, $C_{\text{IV}} \leq 2^{90}$.

Proof. Split U_X, U_Y into the pieces of their h -cells and expand as in Step 1 of Estimate 7.6. A term is *near* if $|c_P - c_{P'}| \leq 4 \cdot 2^{-h}$ and $|c_Q - c_{Q'}| \leq 4 \cdot 2^{-h}$, and *far* otherwise.

Near terms. Each row has at most 81^2 near columns and vice versa; Schur's test as in Step 4 of Estimate 7.6 bounds the near terms by the first sum.

Far terms: the setting. For points $x \in P$, $x' \in P'$, $y \in Q$, $y' \in Q'$ write $d_x = |x - x'|$, $d_y = |y - y'|$, $d_+ = \max(d_x, d_y)$, $d_- = \min(d_x, d_y)$. In a far term $d_+ > (4 - \sqrt{2})2^{-h} > 2.5 \cdot 2^{-h}$. The term equals the integral over the four points of $\text{avg}_{r \approx 2^v} r^2 I_x(r) I_y(r)$, where

$$I_x(r) = \int_{S^1} e^{-irw \cdot (x-x')} a_x(w) dw, \quad a_x(w) = b(x, w)b(x', w),$$

and similarly I_y . Let $M = M_{\mathcal{L}}$; then a_x is $(1, 2M)$ -regular up to order $6T$. Since $2^{h-v} = 4^{-\max(L, \varepsilon N)}$, we have $\max(1, 4^L R^{-2\varepsilon}) 2^{h-v} \leq R^{-2\varepsilon}$, hence

$$M^2 2^{h-v} \leq 2^{14} T^4 (N^2 + 1)^2 R^{-2\varepsilon}, \quad \frac{10^8 (T+1)^4 (2M)^2}{r d} \leq R^{-15\varepsilon/8} \quad \text{for } r \geq 2^{v-2}, d \geq 1.25 \cdot 2^{-h}, \quad (7.5)$$

by (P1). For $d_x \geq 1.25 \cdot 2^{-h}$, Lemma 3.7 with $\Lambda = rd_x$, ϕ_0 the angle of $x - x'$, gives

$$I_x(r) = \left(\frac{2\pi}{rd_x}\right)^{1/2} \sum_{\pm} e^{\mp i(rd_x - \pi/4)} \sum_{j < T} (rd_x)^{-j} c_j^{\pm} + E_x(r),$$

where $c_j^+ = (P_j a_x)(\phi_0)$ and $c_j^- = (\bar{P}_j a_x)(\phi_0 + \pi)$, with $|c_j^{\pm}| \leq \|P_j\|_{2M} \leq (400jM^2)^j$ and $|E_x(r)| \leq R^{-15\epsilon T/8} \cdot 10^8(T+1)^4(2M)^2 \leq R^{-400}$. If $c_j^{\pm} \neq 0$ for some j , then some derivative of a_x is nonzero at $\pm e(x, x')$, so by (S3) and (S2) the direction $e(x, x')$ passes every test of $\mathcal{L}$ at x and at x' at width λ_{i+1} (the tests are even). Write $\mathbf{1}_{\text{pass}}(x, x')$ for the indicator of this event.

Case A: $d_+ > 2d_-$. Say $d_x = d_+$; expand I_x only. In each main term the r -phase is $-r(\pm d_x + w_2 \cdot (y - y'))$, whose derivative has absolute value at least $d_x - d_y \geq d_x/2 \geq 1.25 \cdot 2^{-h}$, uniformly in w_2 . The amplitude $2^{-v} \Psi(r/2^v) r^{3/2-j}$ satisfies $|\partial_r^k(\cdot)| \leq 2^{v/2+3} 4^j 2^{-vj} (600T^2 2^{-v})^k$ for $k \leq 6T$ (on $[2^{v-2}, 2^{v+2}]$, $|\partial^k r^\alpha| \leq (8T)^k 2^{(2-v)k} r^\alpha$ for $|\alpha| \leq 2T$, $k \leq 6T$). The linear-phase case of Lemma 3.4 with $k = T$ and $M' = 600T^2 2^{-v} < 1$, then $|c_j^{\pm}| d_x^{-1/2-j} \leq 2^{h/2} (400jM^2 \cdot 2^h)^j$ and (7.5), bound each of the $2T$ main terms by $2^{2v+10} R^{-15\epsilon T/8} \leq R^{-400}$. The remainder contributes at most $64 \cdot 4^v \cdot 2\pi R^{-400}$.

Case B: $d_+ \leq 2d_-$. Both $d_x, d_y \geq 1.25 \cdot 2^{-h}$; expand both. Products involving a remainder are at most R^{-390} after averaging. Products of main terms with equal signs have the linear r -phase $\mp r(d_x + d_y)$ with $d_x + d_y \geq 2.5 \cdot 2^{-h}$, and are at most R^{-400} each, by the linear-phase case of Lemma 3.4 as in Case A. The products with opposite signs equal

$$\frac{2\pi}{(d_x d_y)^{1/2}} \sum_{j, j' < T} c_j^{\pm} c_{j'}^{\mp} d_x^{-j} d_y^{-j'} 2^{-v} \int \Psi(r/2^v) r^{1-j-j'} e^{\mp i r(d_x - d_y)} dr.$$

Write $m = 1 - j - j'$. On $[2^{v-2}, 2^{v+2}]$ we have $r^m \leq 4^{|m|} 2^{vm}$ and $r^{-1} \leq 4 \cdot 2^{-v}$. Integrating by parts twice, using only $|\Psi| \leq 14$, $|\Psi''| \leq 784$, the r -integral (with its factor 2^{-v}) is at most $4 \cdot 784 (|m| + 1)^2 4^{|m|} 2^{vm} \min(1, (2^v |d_x - d_y|)^{-2}) \leq 12544 (j + j' + 2)^2 4^{j+j'+1} 2^{v(1-j-j')} (1 + 2^v |d_x - d_y|)^{-2}$. By $|c_j^{\pm}| \leq (400jM^2)^j$, $d_x, d_y \geq 1.25 \cdot 2^{-h}$ and (7.5), $|c_j^{\pm}| (4 \cdot 2^{-v} / d_x)^j \leq (1280 T M^2 2^{h-v})^j \leq 4^{-j}$, and similarly for y . There are two opposite-sign products, and $2 \sum_{j, j' \geq 0} (j + j' + 2)^2 4^{-j-j'} = 256/9 < 100$. All coefficients vanish unless both pass conditions hold. So the opposite-sign products together are at most

$$C 2^v (d_x d_y)^{-1/2} (1 + 2^v |d_x - d_y|)^{-2} \mathbf{1}_{\text{pass}}(x, x') \mathbf{1}_{\text{pass}}(y, y'), \quad C = 2\pi \cdot 12544 \cdot 4 \cdot 100 < 2^{25}. \quad (7.6)$$

Regrouping. In Case B let n be the depth with $4K2^{-n} < d_+ \leq 8K2^{-n}$. Then $d_x, d_y \in (2K2^{-n}, 8K2^{-n}]$, so the n -cells $P \ni x$, $P' \ni x'$ are separated, and so are $Q \ni y$, $Q' \ni y'$; moreover $a < n < h + 12$, because $d_+ \leq \sqrt{2} 2^{-a}$ and $d_+ > 2.5 \cdot 2^{-h}$. Let $\eta_{PP'}$ be the image of $(\rho|_P \times \rho|_{P'}) \mathbf{1}_{\text{pass}}$ under $(x, x') \mapsto |x - x'|$. By (7.6), $(d_x d_y)^{-1/2} \leq 2^n / (2K)$ and Lemma 7.3(iii), the contribution of depth n is at most

$$\frac{10C}{2K} 2^{v+n} \sum_{(P, P'), (Q, Q')} ((\eta_{PP'} \times \eta_{PP'}) \{|s - s'| \leq 2^{-v}\}) \cdot (\eta_{QQ'} \times \eta_{QQ'}) \{|s - s'| \leq 2^{-v}\})^{1/2}.$$

If a pair passes $\mathcal{L}$ at width λ_{i+1} , it passes the sublist $\mathcal{L}_n$ at width λ_{i+1} ; and $|x - x'|^{-1/2} \geq (11K2^{-n})^{-1/2}$. Hence $(\eta_{PP'} \times \eta_{PP'}) \{|s - s'| \leq 2^{-v}\} \leq 11K 2^{-n-v} \rho(P) \rho(P') D_{P, P'}(v)$, and the contribution is at most

$$55C \max D(v) \cdot \left(\sum_{(P, P')} (\rho(P) \rho(P'))^{1/2} \right) \left(\sum_{(Q, Q')} (\rho(Q) \rho(Q'))^{1/2} \right) \leq C_{\text{IV}} \max D(v) \rho(X) \rho(Y),$$

since each n -cell has at most $(20K + 1)^2$ separated partners and $(\rho(P)\rho(P'))^{1/2} \leq \frac{1}{2}(\rho(P) + \rho(P'))$. Here $C_{IV} = 55C(20K + 1)^4 < 2^6 \cdot 2^{25} \cdot 2^{58} < 2^{90}$. The negligible parts total at most $R^{-380}\rho(X)^2\rho(Y)^2 \leq R^{-30}\rho(X)\rho(Y)$. Divide by $\rho(X)\rho(Y)$. $\square$

8 The induction for one piece

In this section all cells are cells of one regular piece ρ , with excess function A , and all tests are tests of ρ . The combinatorics is that of [7, §6]; the block length $q = \kappa N$ now depends on N , and every constant is explicit. Since the proofs of the schedule and budget lemmas use only q , A and Lemma 5.10(ii), they are repeated here only briefly.

8.1 Budget and schedule

An interval of depths $J = [a, t]$ is *long* if $t - a > 100q$ and *short* otherwise. If $t - a \geq 10q$ put $J^- = [a, t - 10q]$ and $J^+ = [a + 10q, t]$.

Definition 8.1 (budget). For an interval J of length at least $10q$, $V(J) = \min_{J^-} A + \min_{J^+} A$.

By Lemma 5.10(iii),(iv), $-2\kappa \leq V(J) \leq 2$.

Definition 8.2 (schedule). Cut $\{0, \dots, N\}$ into consecutive blocks of q depths, and in each block *mark* one depth where A is smallest on that block. For a long interval $J = [a, t]$:

- its *inner interval* $I(J)$ is the union of the blocks contained in $[a + q, t - q]$;
- its *split point* $p = p(J)$ is a marked depth in $I(J)$ where A is smallest (ties broken by a fixed rule);
- J is *left-split* if $2p \leq a + t$ and *right-split* otherwise;
- J *contributes* the tube tests $[g, p]$ for $p - \ell \leq g < p$, where $\ell = \min(p - a, t - p)$, and, if J is right-split, also the projection test $[p, t]$.

The list $\mathcal{T}(J)$ consists of all tests contributed by all long intervals $J' \subseteq J$.

There are at most $1/\kappa + 2$ blocks, hence at most $1/\kappa + 2$ marked depths.

Lemma 8.3 (properties of the schedule). *Let $J = [a, t]$ be long and $p = p(J)$.*

- $I(J)$ is an interval containing p and $[a + 2q, t - 2q]$, and $A(n) \geq A(p)$ for every $n \in I(J)$.*
- If $J' \subseteq J$ then $\mathcal{T}(J') \subseteq \mathcal{T}(J)$. All anchors are marked depths, and $\mathcal{T}(J)$ has at most N^2 tests.*
- A test of $\mathcal{T}(J)$ with anchor p' and length ℓ' was contributed by a long interval $J' \subseteq J$, its origin, with $[p' - \ell', p' + \ell'] \subseteq J'$ and $p' = p(J')$; the origin also contributes every tube test $[g, p']$ with $p' - \ell' \leq g < p'$. In particular $\ell' \leq (t - a)/2$.*

Proof. (a) The blocks inside $[a + q, t - q]$ are consecutive and cover it except for less than one block at each end; on each such block A is at least its value at the block's marked depth, which is at least $A(p)$. (b) Immediate; a test is determined by its type, its anchor (at most $1/\kappa + 2$ choices) and its other endpoint (at most $N + 1$ choices), and $2(1/\kappa + 2)(N + 1) \leq N^2$ by (P4). (c) For a tube test $[g, p']$ from $J' = [a', t']$, $\ell' = p' - g \leq \min(p' - a', t' - p')$; for the projection test $[p', t']$ of a right-split J' , $\ell' = t' - p' < p' - a'$. $\square$

8.2 States and moves

Definition 8.4 (states). Fix a level i .

- A D -state is $D_{X,Y}(t)$ for separated a -cells X, Y , computed with the list $\mathcal{T}([a, t])$ at level i . Its interval is $[a, t]$. We write $D[a, t]$ for the maximum over all such pairs of cells.
- An F -state is $F_{X,Y}(v)$ for a -cells X, Y (any two), computed with symbols of order at most $2T$ built at level i from those tests of $\mathcal{T}(B)$ whose anchors are deeper than a . Here $B \supseteq [a, v]$ is an interval ending at most εN after v , the *base*. Its interval is $[a, v]$. We write $F_B[a, v]$ for the supremum over all such pairs of cells and symbols.

A state is long or short according to its interval, and a is its *start*.

Proposition 8.5 (moves). *Every long state at level $i \leq I_* - 2$ is bounded by a sum of at most $N + 13$ terms of the form multiplier $\times$ (value of a child state), plus R^{-25} . The children and multipliers are listed in Table 1. Every child's interval is contained in its parent's interval, and children are at level i or $i + 1$.*

move	applies to	children	multiplier	cost
1. linearize	$D[a, t]$, right-split	$D[a, p]$	$R^{k+7\varepsilon}$	$k = k_p[p, t]$
2. go to the Fourier side	$D[a, t]$, left-split	$D[a, t - \varepsilon N]$	C_1	0
		$F_{[a,t]}[p, v], t - \varepsilon N < v \leq t$	$R^{k+13\varepsilon}$	$k = k_p[a, p]$
3. refine at a long mask	$F_B[a, v], L \geq (v - a)/2$	$F_B[p, v]$	$R^{k+12\varepsilon}$	$k \leq k_p[a, p]$
4. split in space	$F_B[a, v], L < (v - a)/2$	$F_B[h, v]$	81^2	0
		$D[n, v], a < n < h + 12$	C_{IV}	0

Table 1: The four moves. In Moves 1–2, p is the split point of $[a, t]$. In Moves 3–4, L is the largest length of a test still present in the F -state ($L := 0$ if no test is present); in Move 3, p is the anchor of such a longest test; in Move 4, $h = v - 2 \max(L, \varepsilon N)$.

Proof. Each move is one or two of the estimates of §7. Intervals of long states have length more than $100q \geq 100\varepsilon N$.

Move 1. $J = [a, t]$ is right-split, so it contributed the projection test $[p, t]$, and $2p > a + t$. The tests of $\mathcal{T}([a, p])$ are anchored in $[a, p]$ and have lengths at most $(p - a)/2$ (Lemma 8.3(c)), and $\mathcal{T}([a, p]) \subseteq \mathcal{T}([a, t])$. Estimate 7.7 gives the child $D[a, p]$ at level $i + 1$.

Move 2. Apply Lemma 7.4. Its first term is the child $D[a, t - \varepsilon N]$ at level $i + 1$, after dropping the tests not in $\mathcal{T}([a, t - \varepsilon N])$ (positivity). For a high shell $t - \varepsilon N < v \leq t$: every test has length $\ell \leq (t - a)/2$, so $2\ell < v - a + \varepsilon N$, and $v \geq a + 10\varepsilon N$; Estimate 7.5 bounds the shell by $2\rho(X)\rho(Y) \sup F_{X,Y}(v) + R^{-30}\rho(X)\rho(Y)$, with symbols of order at most $2T$ built from $\mathcal{T}([a, t])$ at level i . The interval is left-split, so it contributed the tube test $[a, p]$; apply Estimate 7.6 from a to p with $g = a$. Its hypotheses hold: $v - p \geq (t - a)/2 - \varepsilon N \geq 10\varepsilon N$; $p - a \leq t - p \leq v - p + \varepsilon N$; and every length is at most $(t - a)/2 \leq t - p \leq v - p + \varepsilon N$. Drop the masks reached at depth p ; what remains are F -states with base $[a, t]$, and $t < v + \varepsilon N$. The multiplier is $2C_1 \cdot R^{k_p[a,p]+12\varepsilon} \leq R^{k_p[a,p]+13\varepsilon}$, since $2C_1 = 640 \leq R^\varepsilon$ by (P1), and the errors add up to at most $\varepsilon N C_1 (R^{-30} + 2R^{-80}) \leq R^{-25}$.

Move 3. Let the longest remaining test have length $L \geq (v - a)/2$ and anchor $p > a$. By Lemma 8.3(c) it has an origin $[a', t'] \subseteq B$ with $[p - L, p + L] \subseteq [a', t']$, and the origin contributes the tube tests $[g, p]$ for $p - L \leq g < p$. Since t' is at most the end of B , $p + L \leq v + \varepsilon N$. Take $g = \max(a, p - L)$; since $p - L \leq v + \varepsilon N - 2L \leq a + \varepsilon N$, we have $a \leq g \leq a + \varepsilon N$, and this test is anchored deeper than a , so its mask is present. The hypotheses of Estimate 7.6 hold: every length is at most $L \leq v - p + \varepsilon N$; $p - a \leq L + \varepsilon N \leq v - p + 2\varepsilon N$; and $v - p \geq L - \varepsilon N \geq 10\varepsilon N$. The cost is $k_p[g, p] \leq k_p[a, p]$. Drop the masks reached at depth p .

Move 4. Every remaining test is shorter than $(v - a)/2$, so $h = v - 2\max(L, \varepsilon N) > a$. Estimate 7.8 gives the near child $F_B[h, v]$ (after dropping the masks reached at depth h) and the far children $D[n, v]$ at level $i + 1$. For the latter, $[n, v] \subseteq B$, so $\mathcal{T}([n, v]) \subseteq \mathcal{T}(B)$, and the tests of $\mathcal{T}([n, v])$ are anchored at split points of subintervals of $[n, v]$, hence deeper than $n > a$; so they are among the present masks and may be taken as the sublist.

Move 2 has at most $\varepsilon N + 1$ children and Move 4 at most $N + 12$. $\square$

8.3 The budget pays the costs

Lemma 8.6 (the budget pays). *Let a move go from a long interval J to an interval J' , with cost k .*

- (a) *If $k > 0$, then J' has length at least $40q$ and $k \leq V(J') - V(J)$.*
- (b) *If $k = 0$ and J' has length at least $10q$, then $V(J') \geq V(J)$.*
- (c) *For every interval J'' of length at least $10q$ and every $n \in J''$, $V(J'') \leq 2A(n) + 22\kappa$.*

Proof. This is [7, Lemma 6.5]; we recall the argument. (b) If $J' \subseteq J$ then $J'^{\pm} \subseteq J^{\pm}$. (c) J''^- and J''^+ each contain a depth within $10q$ of n , and A changes by at most $10q/N = 10\kappa$ over $10q$ depths.

(a) *Move 1:* $J' = [a, p]$ with $2p > a + t$, so J' has length more than $50q$, and $k = A(p) - \min_{[p, t]} A$. By Lemma 8.3(a), $A \geq A(p)$ on $[a + 2q, p] \supseteq J'^+$; so $\min_{J'^+} A = A(p)$ and $\min_{J^+} A = \min_{[p, t]} A$, and the right budget rises by exactly k , while the left one does not decrease. *Move 2:* $J' = [p, v]$ with $2p \leq a + t$ and $v > t - \varepsilon N$, so J' has length at least $(t - a)/2 - \varepsilon N > 40q$, and $k = A(p) - \min_{[a, p]} A$. Now $A \geq A(p)$ on $[p, t - 2q] \supseteq J'^-$ and on $[p, t - 10q]$, so $\min_{J'^-} A = A(p)$ and $\min_{J^-} A = \min_{[a, p]} A$: the left budget rises by exactly k . The right budget does not decrease, because $J'^+ = [p + 10q, v] \subseteq [a + 10q, t] = J^+$. *Move 3:* $J = [a, v]$, $J' = [p, v]$ of length at least $L - \varepsilon N > 40q$, and $k \leq A(p) - \min_{[a, p]} A$. If this is positive, let $m \in [a, p]$ with $A(m) = \min_{[a, p]} A < A(p)$, and let $[a', t']$ be the origin of the longest test. $A \geq A(p)$ on its inner interval, which contains $[a' + 2q, p]$, so $m < a' + 2q$, and then $v \leq a + 2L \leq m + 2L < a' + 2q + 2L \leq p + L + 2q \leq t' + 2q$. Hence $J'^- = [p, v - 10q]$ lies in the inner interval of the origin, $\min_{J'^-} A = A(p)$, while $\min_{J^-} A \leq A(m)$ (note $m \leq p \leq v - 40q$, so $m \in J^-$). So the left budget rises by at least k , and the right budget does not decrease, because $J'^+ = [p + 10q, v] \subseteq [a + 10q, v] = J^+$. $\square$

8.4 Chains are short

Lemma 8.7 (chains are short). *Follow any chain of moves (parent, child, grandchild, ...). It contains at most $7/\kappa$ moves other than the low-frequency children of Move 2, and at most $1/\varepsilon$ of those.*

Proof. Along a chain the start never decreases and the end never increases. (1) The high-frequency children of Move 2 and the children of Move 3 move the start to a strictly larger marked depth: at most $1/\kappa + 2$ moves. (2) Move 1 lowers the end by at least q : at most $1/\kappa$ moves. (3) A low-frequency child lowers the end by εN : at most $1/\varepsilon$ moves. (4) A run of consecutive F -states ends with at most one far child of Move 4, and every run except the first begins with a high-frequency child of Move 2: at most $1/\kappa + 3$ far children. (5) A near child of Move 4 either moves the start past the anchor of a longest test (charge it to that marked depth; each is passed once), or keeps the same $L \geq \varepsilon N$ with start $h = v - 2L$, so that the next move, if any, is Move 3 (charge it to that move), or is short (the chain ends): at most $2(1/\kappa + 2) + 1$ moves. The total of (1), (2), (4), (5) is at most $(1/\kappa + 2) + 1/\kappa + (1/\kappa + 3) + (2/\kappa + 5) = 5/\kappa + 10 \leq 7/\kappa$, as $\kappa \leq 1/5$ by (P4). $\square$

8.5 The energy decay theorem

Theorem 8.8 (energy decay). *Let J be a long interval. Every state with interval J (of either kind, with any base) at level 0 or 1 is at most $R^{-V(J)+300\kappa}$.*

Proof. The tree. Apply Proposition 8.5 repeatedly, stopping at short states. By Lemma 8.7, every path from the root has at most $7/\kappa + 1/\varepsilon$ moves, so the levels stay at most $7/\kappa + 1/\varepsilon + 1 \leq I_* - 2$, the tree is finite, and it has at most $(N + 14)^{7/\kappa + 1/\varepsilon + 1} \leq R^\kappa$ nodes by (P3). Unwinding the inequalities,

$$\text{value of the root} \leq \sum_{\text{leaves } \lambda} \left(\prod_{\text{edges to } \lambda} \text{mult} \right) \text{value}(\lambda) + \sum_{\text{internal nodes } \nu} \left(\prod_{\text{edges to } \nu} \text{mult} \right) R^{-25}.$$

Products of multipliers. Along a path, bound 81^2 and $C_{IV} \leq 2^{90}$ by $R^{13\varepsilon}$ (Remark 2.3). Then a path with $m_1 \leq 7/\kappa$ ordinary moves and $m_2 \leq 1/\varepsilon$ low-frequency moves has $\prod \text{mult} \leq R^{\sum \text{cost}} R^{13\varepsilon m_1} C_1^{m_2}$. Here $R^{13\varepsilon \cdot 7/\kappa} \leq R^\kappa$ because $\varepsilon \leq \kappa^2/(5 \cdot 10^5)$ (P4), and $C_1^{1/\varepsilon} = 320^{1/\varepsilon} \leq R^\kappa$ by (P3) (Remark 2.3). The costs along any path add up to at most $V(J') - V(J) \leq 2 + 2\kappa$ for the last interval J' of length at least $10q$ on the path (Lemma 8.6(a),(b); moves after it have cost 0; and $-2\kappa \leq V \leq 2$). Hence every product is at most $R^{2+4\kappa} \leq R^3$, and the second sum is at most $R^\kappa R^3 R^{-25} \leq R^{-21}$.

Leaves. Let the path end at a short state with start a_m , and let J_* be its last interval of length at least $10q$. By Lemma 8.6, the costs add up to at most $V(J_*) - V(J) \leq 2A(a_m) + 22\kappa - V(J)$. The trivial bounds (7.1) and (7.2) give $\text{value}(\lambda) \leq 2600 R^{200\kappa} R^{-2A(a_m)}$, since the interval has length at most $100q = 100\kappa N$. So each leaf contributes at most

$$R^{2A(a_m)+22\kappa-V(J)} \cdot R^{2\kappa} \cdot 2600 R^{200\kappa-2A(a_m)} \leq 2600 R^{-V(J)+224\kappa},$$

and there are at most R^κ leaves. Altogether the root is at most $2600 R^{-V(J)+225\kappa} + R^{-21} \leq R^{-V(J)+300\kappa}$, using $V(J) \leq 2$ and $2600 \leq R^\kappa$ (by (P3), $\kappa N \geq 80$). $\square$

9 Proof of Theorem 1.1

9.1 The entry depth and the lists

The energy decay theorem gains only where A is large, and $A(0) = 0$. So each piece first jumps, by one application of Estimate II, to a depth after which its excess function stays above β .

Definition 9.1 (entry depth). For a regular piece with excess function A , its *entry depth* is $c = \max\{n : A(n) \leq \beta\}$.

Lemma 9.2 (entry). *For every regular piece of μ_1 or μ_2 , and $N \geq N_0$:*

- (i) $\lfloor \beta N \rfloor \leq c < N/4$, and $[c, N]$ is long;
- (ii) $A(n) > \beta - 1/N$ for all $n \in [c, N]$; hence $V([c, N]) \geq 2\beta - 2/N$;
- (iii) the entry cost $k_c[0, c]$ is at most $\beta + \kappa$.

Proof. (i) By Lemma 5.10(iv), $A(n) \leq n/N \leq \beta$ for $n \leq \beta N$. For $n \geq N/4$, Lemma 5.10(iii) and the monotonicity of g give $A(n) \geq g(N/4) - \kappa = 2\beta - \kappa > \beta$, as $\kappa \leq \beta/500$ (P4). Also $N - c > 3N/4 > 100\kappa N = 100q$. (ii) For $n > c$, $A(n) > \beta$ by maximality, and $A(c) \geq A(c+1) - 1/N$ by Lemma 5.10(ii). Both budgets of $[c, N]$ are minima over subsets of $[c, N]$. (iii) $A(c) \leq \beta$, and $A \geq -\kappa$ everywhere by Lemma 5.10(iii). $\square$

The lists. The list $\mathcal{L}(\rho)$ of a regular piece ρ with entry depth c consists of the tube test $[0, c]$ and the tests of $\mathcal{T}([c, N])$. It has at most $N^2 + 1$ tests and depends only on ρ . These are the lists used in the filtered measure τ_N of Definition 6.12. No test of $\mathcal{T}([c, N])$ is anchored at a depth at most c , since anchors are split points of long subintervals of $[c, N]$ and lie in their inner intervals.

9.2 The shell estimate

Proposition 9.3 (shell estimate for one pair of pieces). *Let ρ_1 be a regular piece of μ_1 and ρ_2 a regular piece of μ_2 , and let ν be the image under $(x, y) \mapsto |x - y|$ of*

$$|x - y|^{-1/2} b(x, e(x, y)) b(y, e(x, y)) d\rho_1(x) d\rho_2(y),$$

with b as in Definition 6.12. Then

$$\int_{R/2 \leq |r| \leq 2R} |\hat{\nu}(r)|^2 dr \leq R^{-\beta/3}.$$

Proof. Step 1: Mattila's identity with masks at the root. Apply Estimate 7.5 in the root case, with $a = 0$, $v = N$ and level 0; both pieces have mass 1. Its hypotheses hold: the tube test $[0, c_j]$ has length $c_j < N/4$, and the tests of $\mathcal{T}([c_j, N])$ have length at most $(N - c_j)/2$, so $2\ell \leq N$. Hence the left side is at most $2 \sup F_{S_1, S_2}(N) + R^{-30}$.

Step 2: Cauchy-Schwarz separates the pieces. In r ,

$$F_{S_1, S_2}(N) = \operatorname{avg}_{r \approx R} \sigma_{S_1}(r) \sigma_{S_2}(r) \leq \left(\operatorname{avg}_{r \approx R} \sigma_{S_1}(r)^2 \right)^{1/2} \left(\operatorname{avg}_{r \approx R} \sigma_{S_2}(r)^2 \right)^{1/2}.$$

The j -th factor is $F_{Q_0, Q_0}(N)^{1/2}$ for the piece ρ_j alone, with both cells equal to the root cell $Q_0 = [0, 1]^2$ and carrying the same symbol, of order at most $2T$, built from $\mathcal{L}(\rho_j)$ at level 0. From now on we treat one piece $\rho = \rho_j$, with excess function A and entry depth c .

Step 3: one orthogonality step to the entry depth. Apply Estimate 7.6 to $X = Y = Q_0$, $a = 0$, $p = c$, $v = N$, with the tube test $[0, c]$ (so $g = 0$). Its hypotheses hold: $N - c \geq 3N/4 \geq 10\epsilon N$; $c \leq N - c$; and all lengths are at most $\max(c, (N - c)/2) \leq N - c$. So

$$F_{Q_0, Q_0}(N) \leq R^{k_c[0, c] + 12\epsilon} \max_{P, Q} F_{P, Q}(N) + R^{-80},$$

the maximum over c -cells P, Q . Drop the masks reached at depth c , among them those of $[0, c]$. What remains are symbols built from the tests of $\mathcal{T}([c, N])$, all anchored deeper than c : so $F_{P,Q}(N)$ is bounded by the F -state $F_{[c,N]}[c, N]$ at level 0, whose interval $[c, N]$ is long.

Step 4: energy decay. By Theorem 8.8, $F_{P,Q}(N) \leq R^{-V([c,N])+300\kappa}$.

Step 5: combine. By Lemma 9.2,

$$F_{Q_0,Q_0}(N) \leq R^{(\beta+\kappa)+12\varepsilon-(2\beta-2/N)+300\kappa} + R^{-80} = R^{-\beta+301\kappa+12\varepsilon+2/N} + R^{-80} \leq 2R^{-0.378\beta},$$

because $301\kappa \leq 0.602\beta$ and $12\varepsilon + 2/N \leq 0.02\beta$ by (P4). This holds for both pieces, so the left side of the proposition is at most $2 \cdot 2R^{-0.378\beta} + R^{-30} \leq 5R^{-0.378\beta} \leq R^{-\beta/3}$, because $R^{0.044\beta} = 2^{0.044\beta N} \geq 5$ (by (P4), $\beta N \geq 200$). $\square$

The whole argument in one line. For each piece, with entry depth c ,

$$\int_{|r| \approx R} |\hat{\nu}|^2 \lesssim \prod_{j=1,2} F_{Q_0,Q_0}^{(j)}(N)^{1/2},$$

$$F_{Q_0,Q_0}(N) \lesssim \underbrace{R^{k_c[0,c]}}_{\leq R^{\beta+\kappa}} \cdot \underbrace{\max_{c\text{-cells}} F(N)}_{\leq R^{-V([c,N])+300\kappa} \leq R^{-2\beta+300\kappa+2/N}} \leq R^{-0.378\beta},$$

up to a factor $R^{12\varepsilon}$. Here $R^\beta = 2^{c_\theta N^\theta}$, and $\kappa \approx \beta/1000$, $\varepsilon \approx \kappa^2/10^6$. The entry jump costs at most $\beta + \kappa$ and the decay after it gains at least 2β , so a net gain of about β survives; the budget losses $R^{300\kappa}$ eat less than two thirds of it.

9.3 Conclusion

Proof of Theorem 1.1. Replace E by the similar copy of Proposition 5.6, and let α be the distance measure of μ_1, μ_2 ; it is nonzero and carried by $\Delta(E)$. For each $N \geq N_0$, give every regular piece its list as above, and form τ_N as in Definition 6.12. By Lemma 6.13, $0 \leq \tau_N \leq \alpha$ and $\alpha(\mathbb{R}) - \tau_N(\mathbb{R}) \leq a_N$ with $\sum_N a_N < \infty$.

Write $\tau_N = \sum m_1 m_2 \nu_{\rho_1 \rho_2}$, where ρ_1, ρ_2 run over the regular pieces of μ_1, μ_2 , m_j is the original mass of the piece ρ_j , and $\nu_{\rho_1 \rho_2}$ is the measure of Proposition 9.3. Since $\sum m_1 m_2 \leq 1$, the triangle inequality in L^2 of the shell and Proposition 9.3 give

$$\int_{2^{N-1} \leq |r| \leq 2^{N+1}} |\hat{\tau}_N(r)|^2 dr \leq R^{-\beta/3} = 2^{-c_\theta N^\theta/3},$$

which is summable in N . By Lemma 4.2, α is absolutely continuous. Since it is nonzero and carried by $\Delta(E)$, $|\Delta(E)| > 0$. $\square$

10 Remarks

Where $\theta > 2/3$ is used. Only in conditions (P2) and (P3). Both ask that a cost of size $2^{O(1/\varepsilon)}$ or $N^{O(1/\kappa+1/\varepsilon)}$ be at most $R^{\kappa/4}$:

- (a) the number of types in the regular decomposition, at most $(48/\varepsilon + 1)10^{8/\varepsilon+1}$ (Lemma 5.8; this is (P2));

- (b) the factor $C_1 = 320$ paid at each of the at most $1/\varepsilon$ low-frequency steps of a chain, $C_1^{1/\varepsilon}$ in total (Lemma 7.4, Theorem 8.8; this is (P3));
- (c) the number of nodes of the tree of moves, at most $(N + 14)^{7/\kappa + 1/\varepsilon + 1}$ (Theorem 8.8; this is (P3)).

With $\varepsilon \asymp \kappa^2$ and $\kappa \asymp N^{\theta-1}$, (P2) reads $\kappa^{-2} \lesssim \kappa N$, that is $N^{3\theta-2} \rightarrow \infty$, and (P3) reads $\kappa^{-2} \log N \lesssim \kappa N$, that is $N^{3\theta-2} \gg \log N$. Both hold for large N exactly when $\theta > 2/3$; at $\theta = 2/3$ the quantity $\kappa^3 N$ stays bounded and both fail. Every other condition holds for $\theta > 1/2$, except the filter as written: Lemma 6.13 uses the Orlicz exponent 8 and needs $8(2\theta - 1) > 1$, that is $\theta > 9/16$. Since Corollary 5.5 gives finite logarithmic energies of every order, the exponent 8 can be replaced by any $\gamma > 1/(2\theta - 1)$, and then the filter works for every $\theta > 1/2$.

How the threshold could be lowered. The three costs above all come from steps whose number is of order $1/\varepsilon$ rather than $1/\kappa$: a type of the regular decomposition records one of ten choices at each of $O(1/\varepsilon)$ checkpoint depths (spaced about $\varepsilon N/4$ apart), and a chain can contain $O(1/\varepsilon)$ low-frequency steps, each lowering the end of the interval by only εN . If each of the three costs were reduced to $2^{O(1/\kappa)} N^{O(1/\kappa)}$, the binding condition would become $\kappa^{-1} \log N \lesssim \kappa N$, that is $\theta > 1/2$. Below $1/2$ the method in its present form stops for a more basic reason: a chain has about $1/\kappa$ ordinary steps, each losing $R^{C\varepsilon}$, so the total loss $R^{C\varepsilon/\kappa}$ is affordable only if $\varepsilon \lesssim \kappa^2 \asymp \beta^2$; and the integrations by parts need $\varepsilon N \gg \log N$ (P1). Together these force $\beta^2 N \gg \log N$, that is $\theta > 1/2$. This is the structural obstruction that [9, §7] identifies for logarithmic gauges, where the available gain is only a power of $\log R$.

Relation to [9]. The two ingredients taken from the research note are its Orlicz radial estimate (Theorem 1 there, used here with exponent 8; under the gauge h_θ every logarithmic moment is finite, Corollary 5.5) and its summable reconstruction criterion (Lemma 7 there). The note's §7 lists three obstacles for logarithmic gauges: the regularization budget, the recursive loss, and the angular cutoffs. For h_θ with $\theta > 2/3$ they are met as follows.

- *Regularization budget.* The two roles of the single Lean parameter are separated: the regularity tolerance is R^ε , while light parts are discarded at mass $R^{-\kappa/2}$. The number of types is then paid by $R^{\kappa/4}$, which is (P2).
- *Recursive loss.* This is the choice $\kappa \approx \beta/1000$, $\varepsilon \approx \kappa^2/10^6$, together with (P3).
- *Angular cutoffs.* At each point the tests of a list fail on a set of directions of arc length at most $2\pi(N^2 + 1)R^{-2\varepsilon}$, and the mass removed at scale N is at most a constant times $R^{-\kappa/4} + NR^{-\varepsilon} + (\varepsilon N)^{-8}$, which is summable in N . No exponential smallness, as in [9, Lemma 6], is needed, because the criterion (Lemma 4.2) only asks for summability. This works because the per-step factor $R^{2\varepsilon}$ is $2^{O(N^{2\theta-1})}$, far below the gain $2^{c_\theta N^\theta}$.

What is proved here and what is quoted. Everything specific to the problem is proved in these notes. Quoted are standard facts: Faà di Bruno's formula, Cauchy's estimates, Plancherel's theorem, Lévy's continuity theorem, the Dunford–Pettis theorem, Mazur's lemma, uniqueness of disintegrations, and the Fourier transform of $e^{i\Lambda s^2/2}$. The combinatorial lemmas of §8 are those of [7], whose proofs are recalled; the estimates of §7 are proved here in full, including the stationary phase and integration by parts that [7] only sketched. These notes have not been refereed.

A Ledger of the scale-dependent conditions

All conditions below hold for $N \geq N_0(\theta, E)$. The last column says what each condition needs as $N \rightarrow \infty$, with $\beta \asymp \kappa \asymp N^{\theta-1}$ and $\varepsilon \asymp N^{2\theta-2}$.

condition	where it is used	needs
$2^{80}(T+1)^{12}(N+16)^{12} \leq R^{\varepsilon/8}$ (P1)	every “polynomial factor $\leq R^{\varepsilon/8}$ ”; integration by parts gains $R^{-3\varepsilon/8}$ per step; absolute constants $\leq 2^{90}$ are $\leq R^\varepsilon$	$\varepsilon N \asymp N^{2\theta-1} \gg \log N$: $\theta > \frac{1}{2}$
$\varepsilon T \geq 240$	negligible errors $R^{-3\varepsilon T/8} \leq R^{-90}$	definition of T
$\varepsilon N \geq 16, \log_2(49/\varepsilon) + 3.33(8/\varepsilon + 1) \leq \kappa N/4$ (P2)	number of types in the regular decomposition $\leq R^{\kappa/4}$	$\kappa^3 N \asymp N^{3\theta-2} \rightarrow \infty$: $\theta > \frac{2}{3}$
$(7/\kappa + 1/\varepsilon + 2) \log_2(N+16) \leq \kappa N/4$ (P3)	number of nodes of the tree $\leq R^\kappa$; $C_1^{1/\varepsilon} \leq R^\kappa$; $2600 \leq R^\kappa$	$N^{3\theta-2} \gg \log N$: $\theta > \frac{2}{3}$
$\mu(G) \geq R^{-\kappa/2}$ for kept parts	$A(n) \geq g(n) - \kappa$; discarded mass $\leq 2R^{-\kappa/4}$	(P2)
$\log_2 C_0 \leq \kappa N/2$	$A(n) \geq g(n) - \kappa$	$\kappa N \rightarrow \infty$
$\varepsilon \leq \kappa^2/(5 \cdot 10^5)$ (P4)	$R^{13\varepsilon \cdot 7/\kappa} \leq R^\kappa$; $12\varepsilon \leq 0.01\beta$	$1/N = o(\kappa^2)$: $\theta > \frac{1}{2}$
$\kappa \leq \beta/500$ (P4)	$R^{301\kappa} \leq R^{0.602\beta}$ in the shell estimate; $2\beta - \kappa > \beta$	definition of κ and $\beta N \geq 1000$
$2/N \leq \beta/100$ (P4)	$V([c, N]) \geq 2\beta - 2/N$; $R^{0.044\beta} \geq 5$	$\beta N \asymp N^\theta \rightarrow \infty$
$I_* \leq T, 2(1/\kappa + 2)(N+1) \leq N^2$ (P4)	masks exist at all levels; lists have at most N^2 tests	$1/\kappa = o(1/\varepsilon)$, $1/\kappa = o(N)$
$1/I_* \geq 8R^{-\varepsilon}, 4I_* \leq R^{3\varepsilon/2}$	masks exist; rotation by $2^{-\ell} R^{\varepsilon/2}$ costs one level	from (P1), $I_* \leq T$
$\sum_N (\varepsilon N)^{-8} < \infty$	the filter removes summable mass	$8(2\theta - 1) > 1$: $\theta > \frac{9}{16}$
$\sum_N R^{-\beta/3} < \infty$	summable shell energies	$\beta N \asymp N^\theta$

References

- [1] N. Dunford and J. T. Schwartz, *Linear Operators, Part I*, Interscience, 1958 (Dunford–Pettis theorem, IV.8.9).
- [2] K. J. Falconer, *On the Hausdorff dimensions of distance sets*, Mathematika 32 (1985), 206–212. <https://doi.org/10.1112/S0025579300010998>
- [3] L. Guth and N. H. Katz, *On the Erdős distinct distances problem in the plane*, Ann. of Math. 181 (2015), 155–190. <https://arxiv.org/abs/1011.4105>
- [4] L. Hörmander, *The Analysis of Linear Partial Differential Operators I*, Springer, 1983 (§7.6–7.7).
- [5] *falconer-all-dimensions*, Lean formalization extracted from openai/math, revision 4ce0748. <https://github.com/CoolRmal/falconer-all-dimensions>
- [6] P. Mattila, *Geometry of Sets and Measures in Euclidean Spaces*, Cambridge University Press, 1995 (Frostman’s lemma, Theorem 8.8).

- [7] *Falconer's distance problem in the plane: a proof organized around energy per unit mass*, notes in this project (`claude/falconer-2d-clean-proof.tex`), October 8, 2026.
- [8] T. Orponen, *On the dimension and smoothness of radial projections*, Anal. PDE 12 (2019), 1273–1294. <https://arxiv.org/abs/1710.11053>
- [9] *Orlicz radial projections and logarithmic distance gauges*, research note (planar case), October 8, 2026.
- [10] OpenAI, *The Falconer distance conjecture in all dimensions*, preprint, 23 September 2026. <https://github.com/openai/math/blob/ad7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/The-Falconer-distance-conjecture-in-all-dimensions-September-23-2026/paper.pdf>